

DASAR GENETIKA MOLEKULER HEMOGLOBINOPATI

Danus Hermawan
Departemen Biokimia
Fakultas Kedokteran UNS



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I . INTRODUCTION

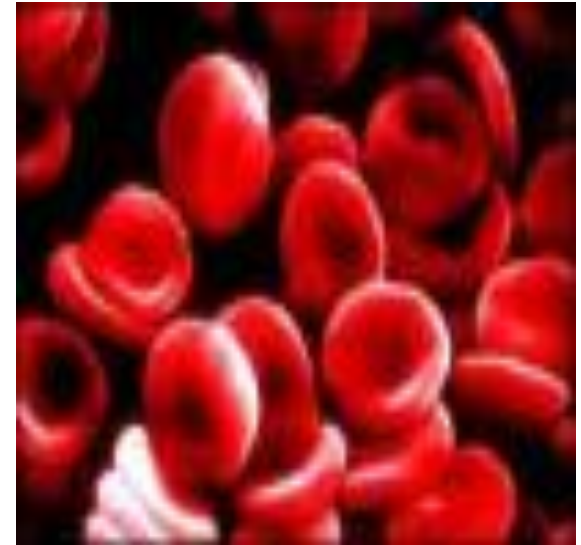
The human hemoglobin (Hb)

- **Each RBC contains Hb**
- **Consist of heme + globin**
- **Function: transports :**
 - **O₂ to the tissue**
 - **CO₂ from the tissue**
- **Anemias is an abnormalities of Hb**
 - ex.: hemolytic anemia**
 - iron deficiency anemia**
 - megaloblastic anemia**

Abnormal Hb with low functionality

Mutation may be deletion, substitution, insertio

- **Haemoglobinopathies are ‘inherited abnormalities of globin chain synthesis’ (*B. Bain 2007*).**



HEMOGLOBINOPATHIES

Hemoglobinopathies occupy a special place in human genetics for many reasons:

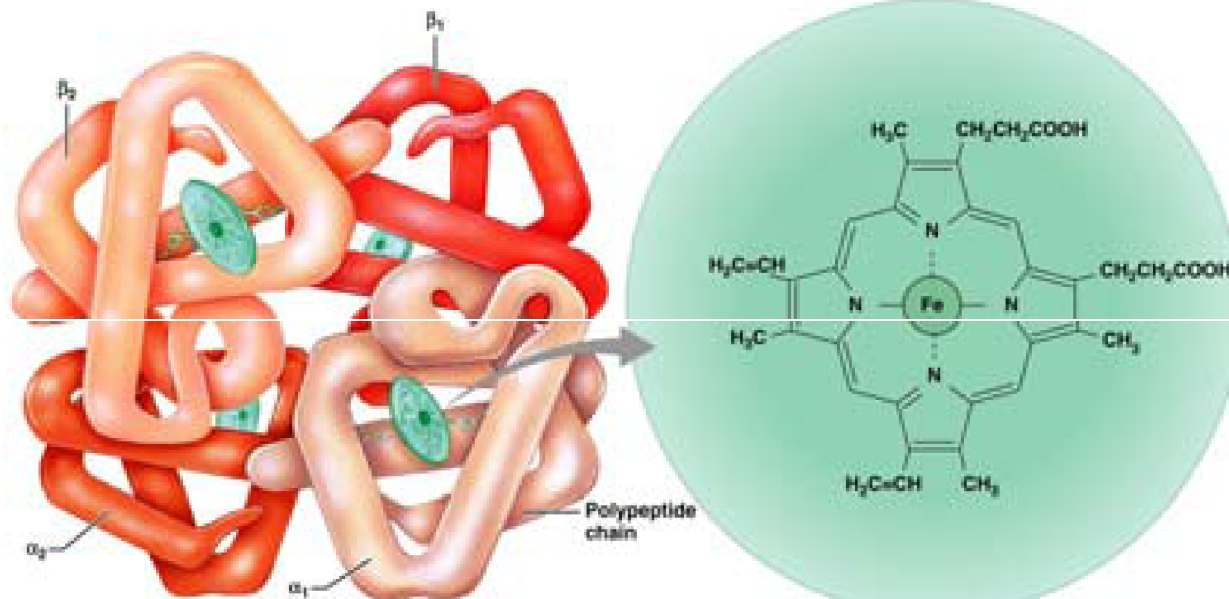
- **They are the most common serious Mendelian diseases**
- **Globins illuminate important aspects of evolution of the genome and of diseases in populations**
- **Developmental controls are probably better understood for globins than for any other human genes**
- **More mutations and more diseases are described for hemoglobins than for any other gene family**
- **Clinical symptoms follow very directly from malfunction of the protein, so that the relationship between molecular and clinical events is clearer for the hemoglobinopathies than for most other diseases**



II . STRUCTURE OF HEMOGLOBIN (HB)

A normal Hb molecule consists of:

- **Four heme molecules: porphyrin ring and iron**
 - **Four globin chains (2 alpha and 2 beta).**
- Each globin chain has an iron in the heme molecule**
- **The iron in the heme molecule binds to oxygen.**
 - **The first hb synthesized in the 8 week embryonic**
 - **Globin: - Alpha family maps to chromosome 16**
 - Beta family maps to chromosome 11

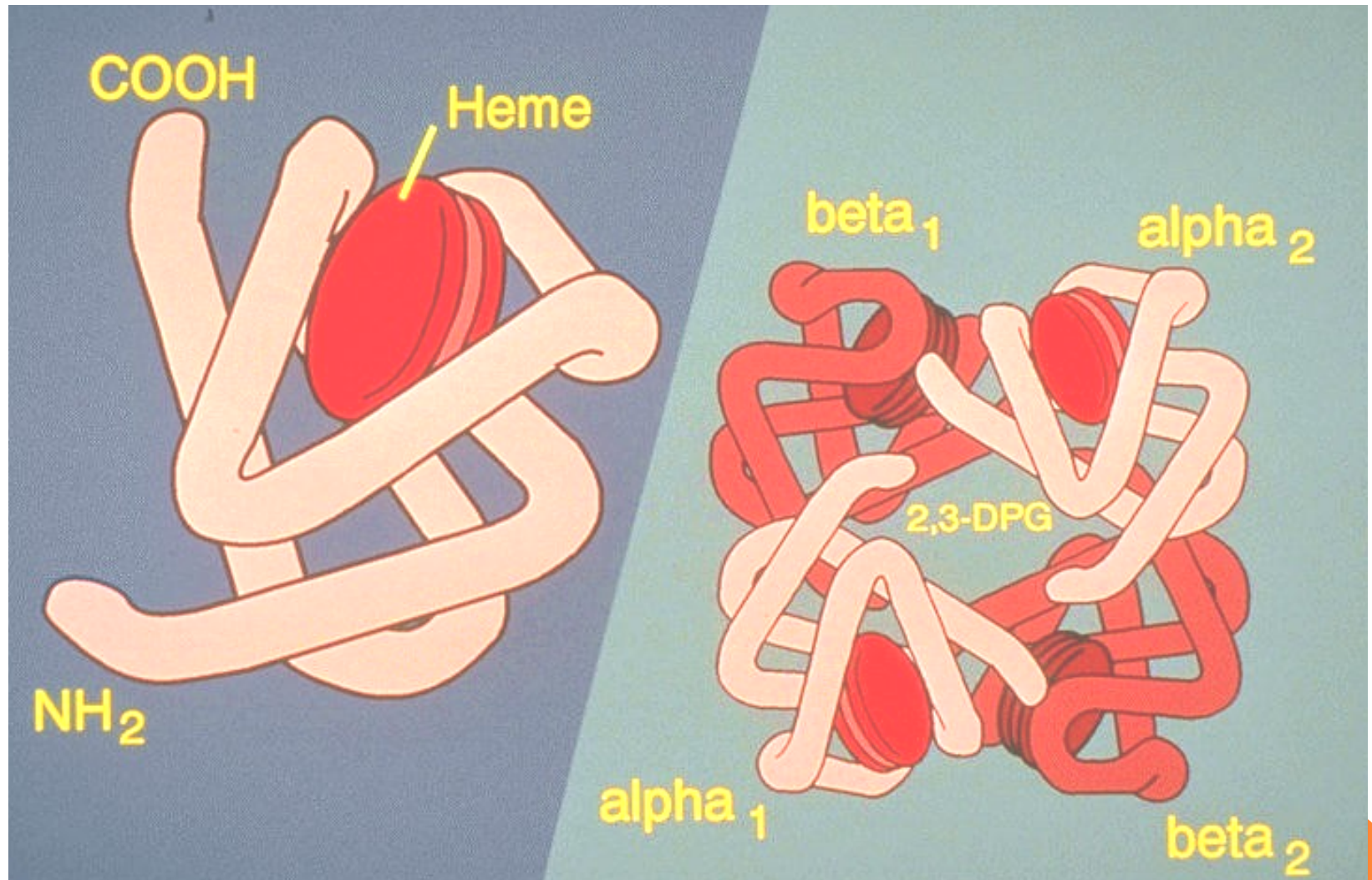


(a) Hemoglobin

(b) Iron-containing heme group

1 UNIT

4 UNITS



III. SYNTHESIS OF HEMOGLOBIN

A. SYNTHESIS OF HEME

Porphyrin-ring

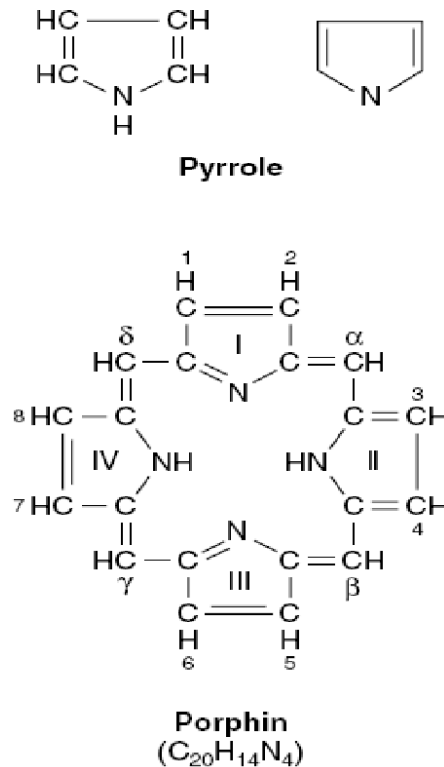
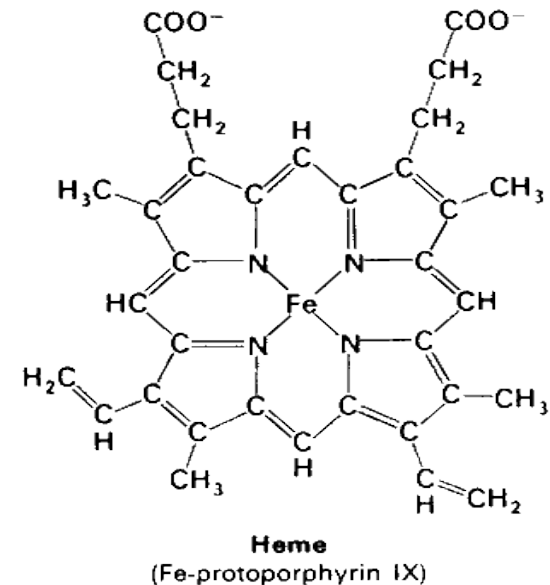


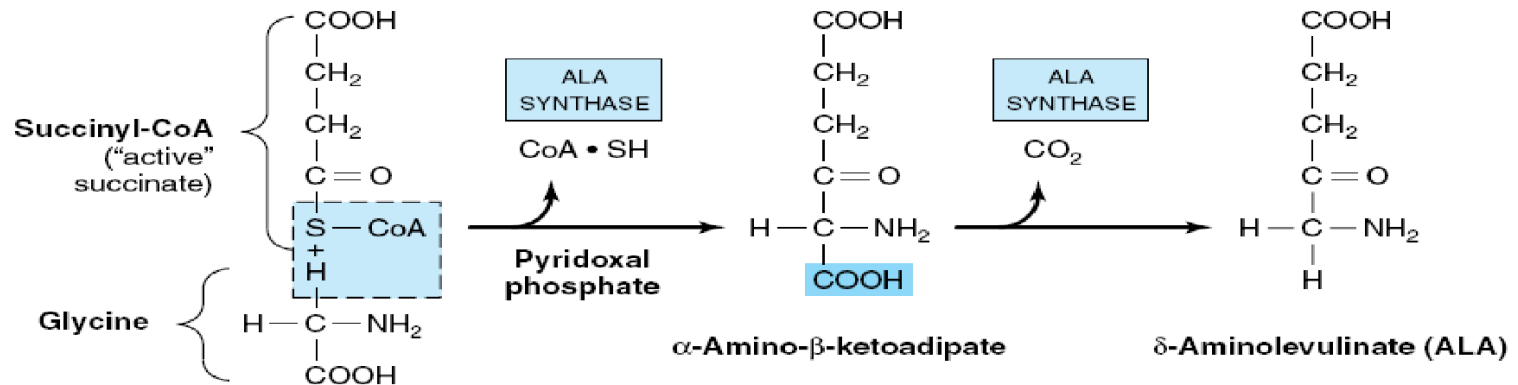
Figure 32-1. The porphin molecule. Rings are labeled I, II, III, and IV. Substituent positions on the rings are labeled 1, 2, 3, 4, 5, 6, 7, and 8. The methenyl bridges (—HC=) are labeled α , β , γ , and δ .

Heme



Iron is coordinated to 4 pyrrole nitrogens of protoporphyrin, and to an nitrogen of a histidine residue from the globin side of the porphyrin.

- Condensation of Succinyl-coA + Glycine $\rightarrow$ ALA (in mitochondria)



- CONDENSATION OF TWO MOLECULES AMLEV $\rightarrow$ PBG (IN CYTOSOL)

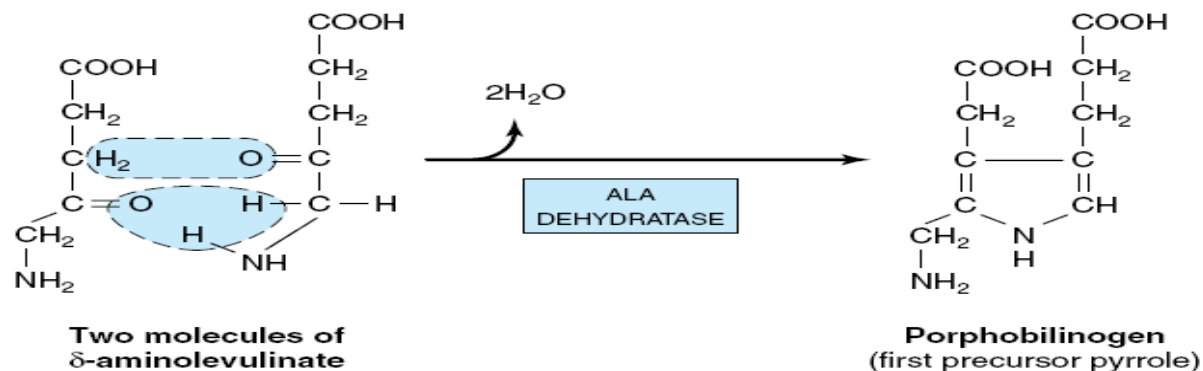


Figure 32-5. Biosynthesis of porphobilinogen. ALA synthase occurs in the mitochondria, whereas ALA dehydratase is present in the cytosol.



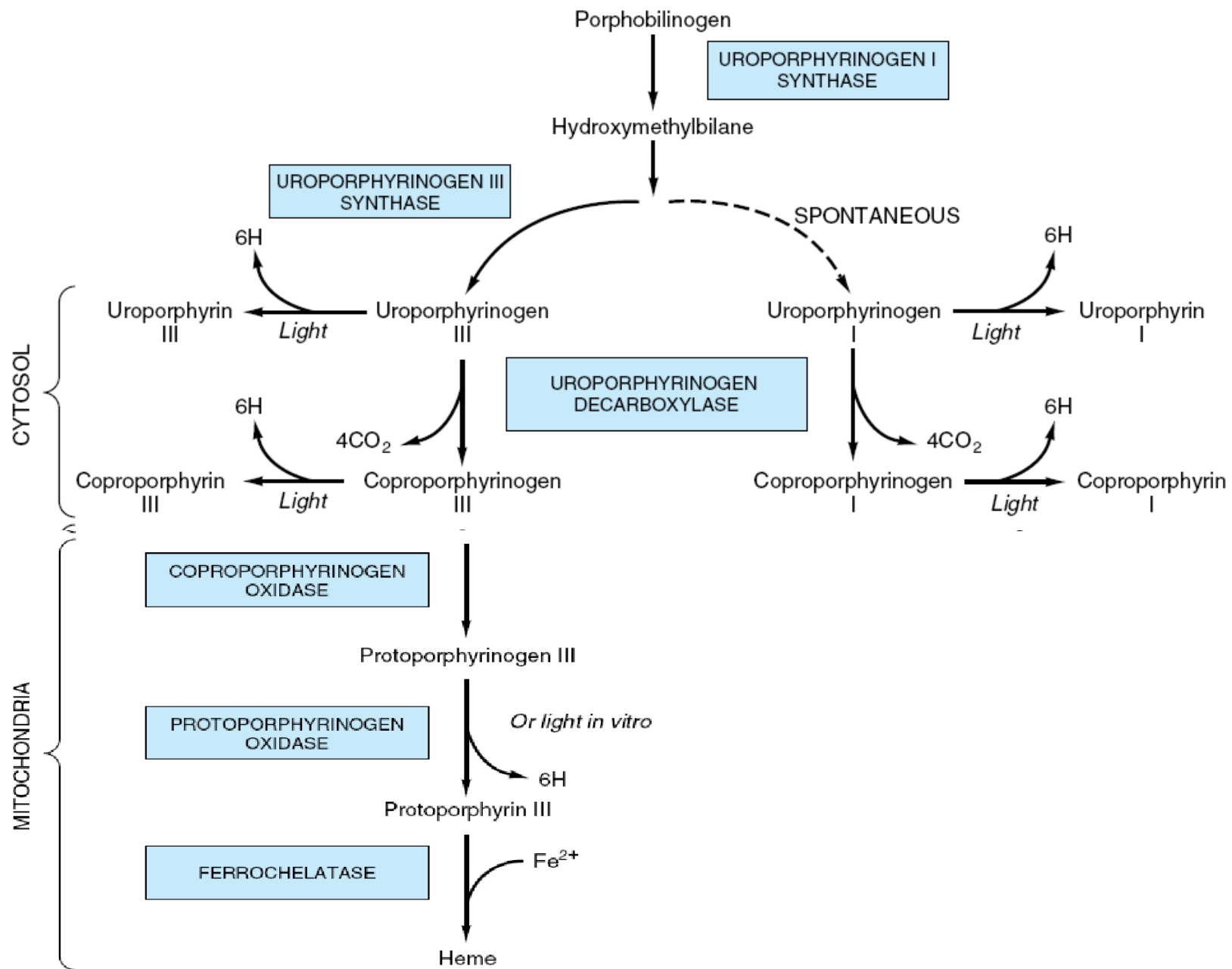
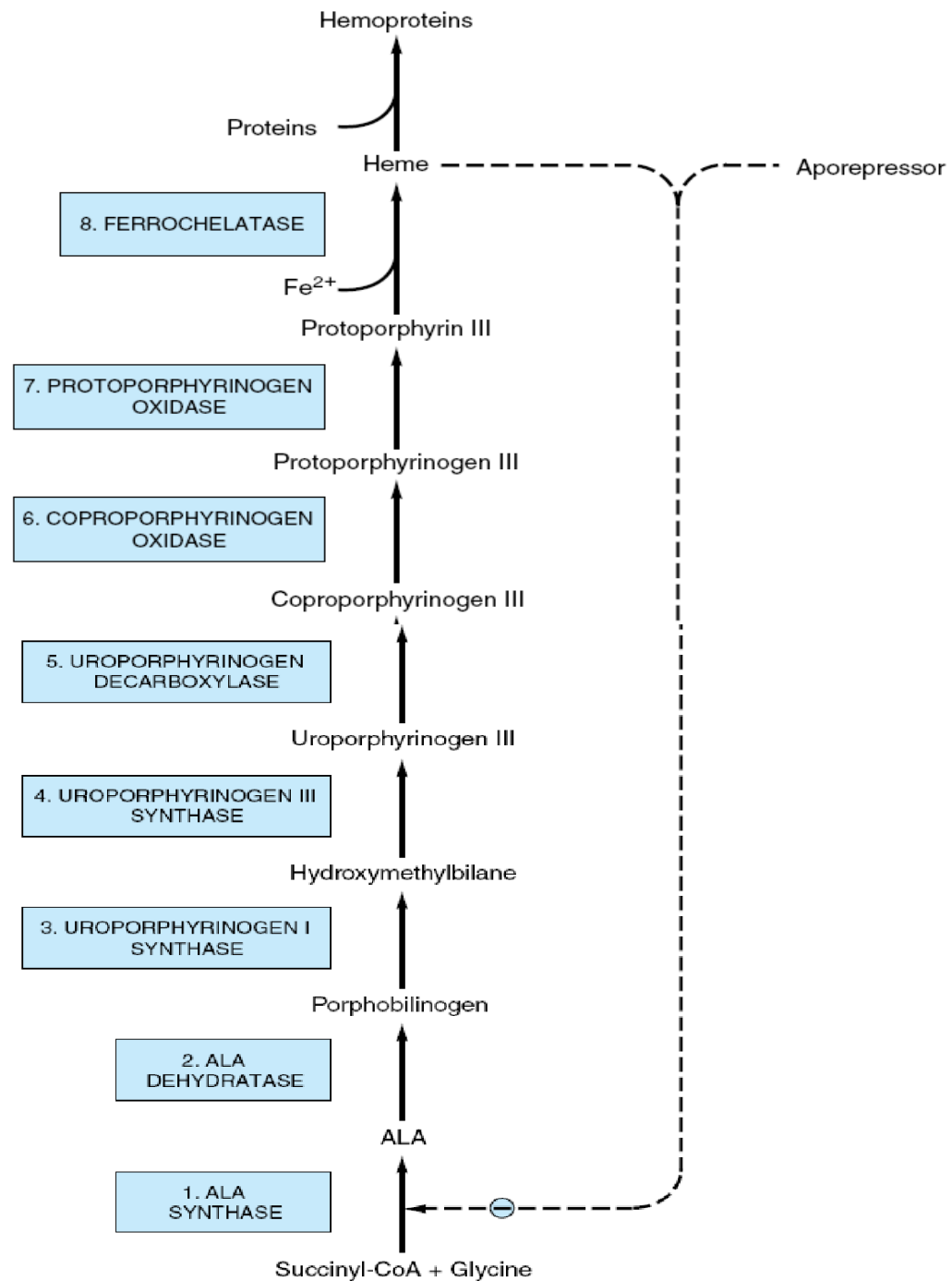


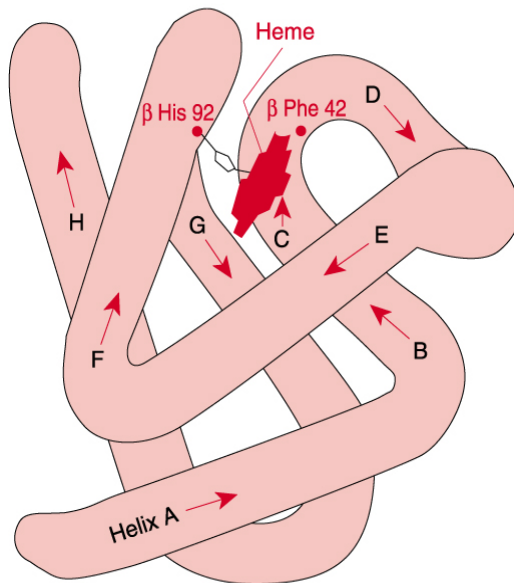
Figure 32–8. Steps in the biosynthesis of the porphyrin derivatives from porphobilinogen. Uroporphyrinogen I synthase is also called porphobilinogen deaminase or hydroxymethylbilane synthase.



B. STRUCTURE OF GLOBIN

Human Hb are composed of 4 peptide chains called **globin**

→ Normal human Hb of the adult is composed of **a pair of two identical chains** (alpha and beta)



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- 8 helices: A-H
- His 92 globins → covalently links heme iron

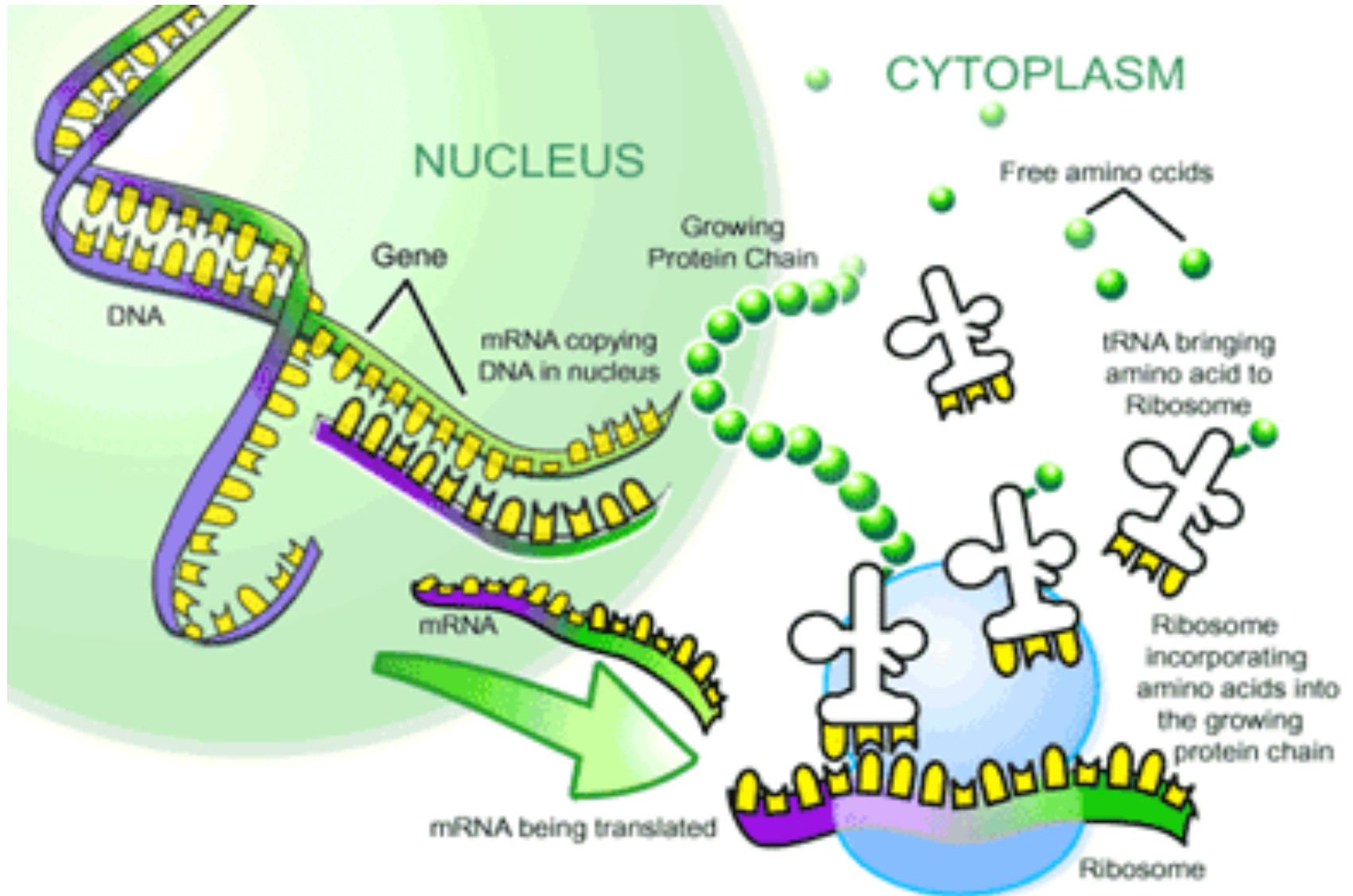


A model of hb:

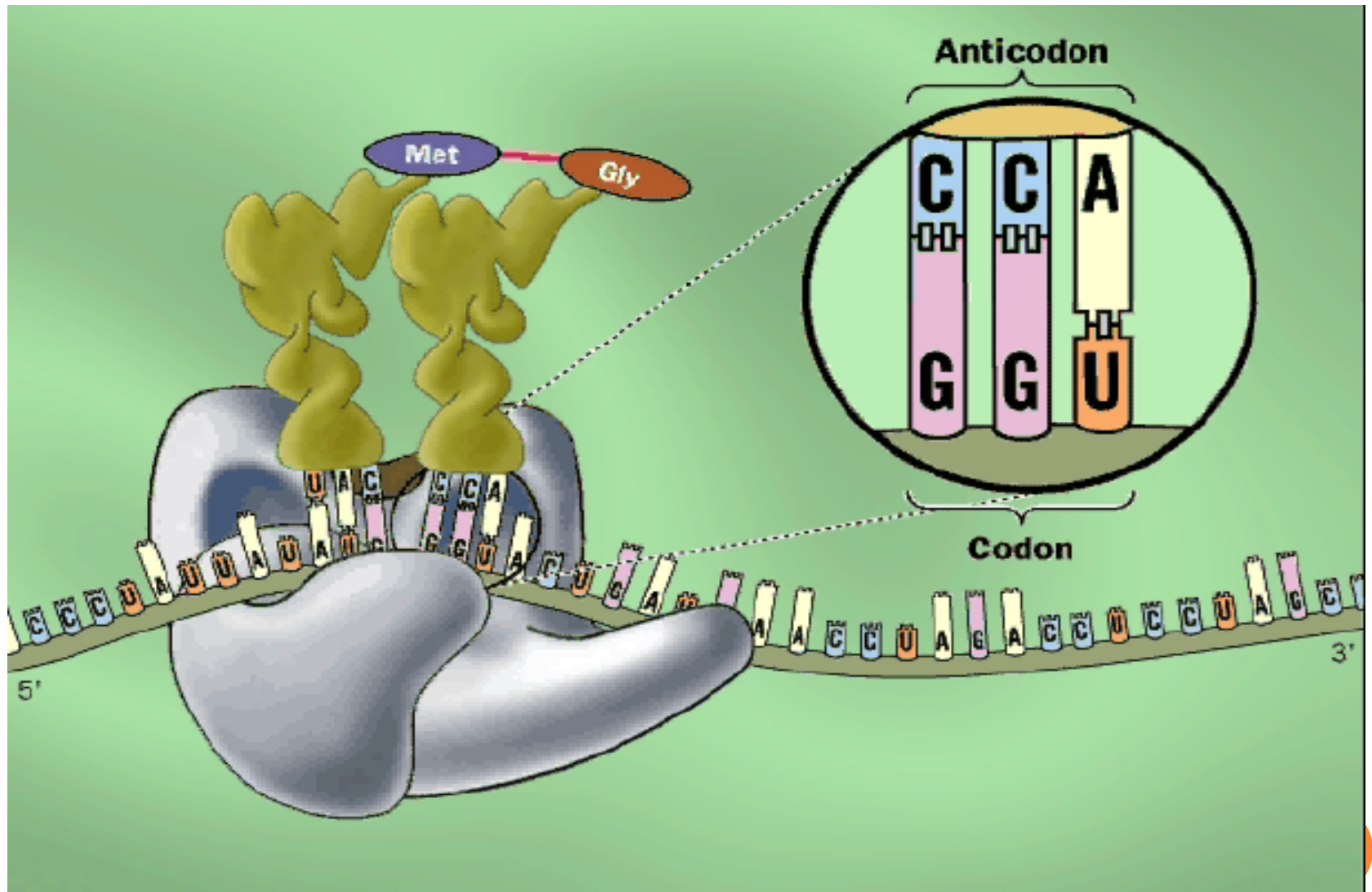
- The alpha chains in this model are **yellow**,
- the beta chains are **pink**,
- the heme groups **blue**



SYNTHESIS OF PROTEIN (GLOBIN)



TRANSLATION



GENETIC CODES (CODON) AND PROTEIN SYNTHESIS

start
codon

mRN
A



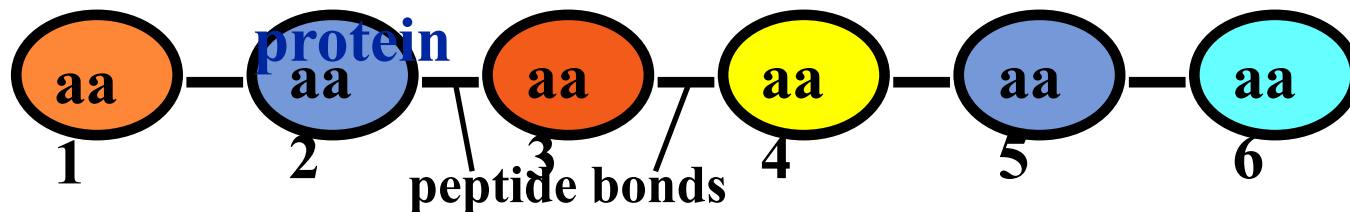
codon 1 codon 2 codon 3 codon 4 codon 5 codon 6 codon 7

protein



<#>

Primary structure of a

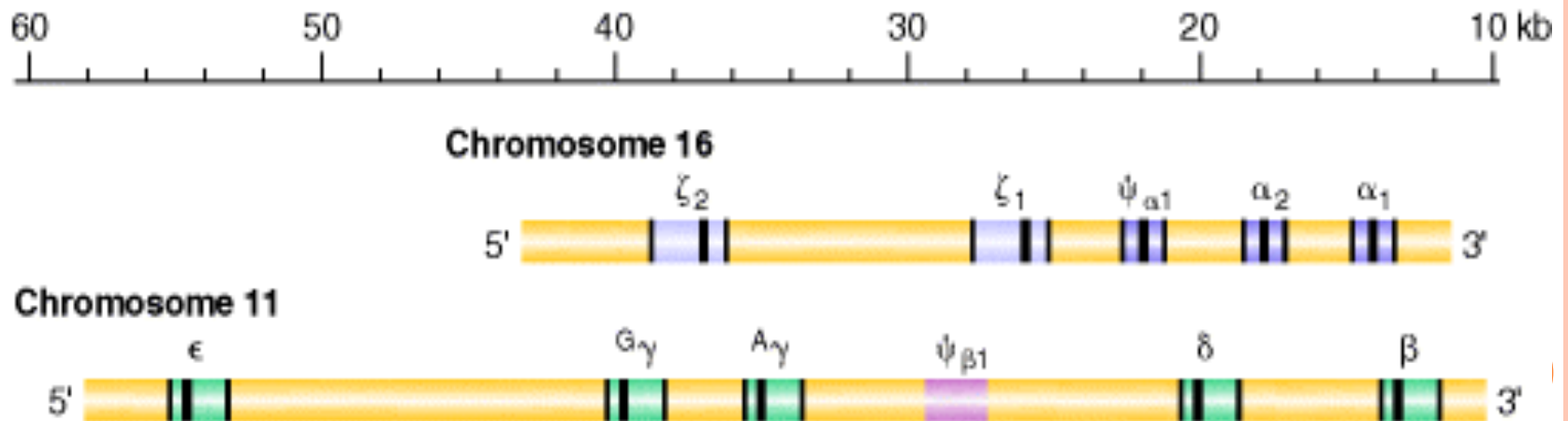


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III. CHROMOSOMAL LOCATION OF GLOBIN GENES

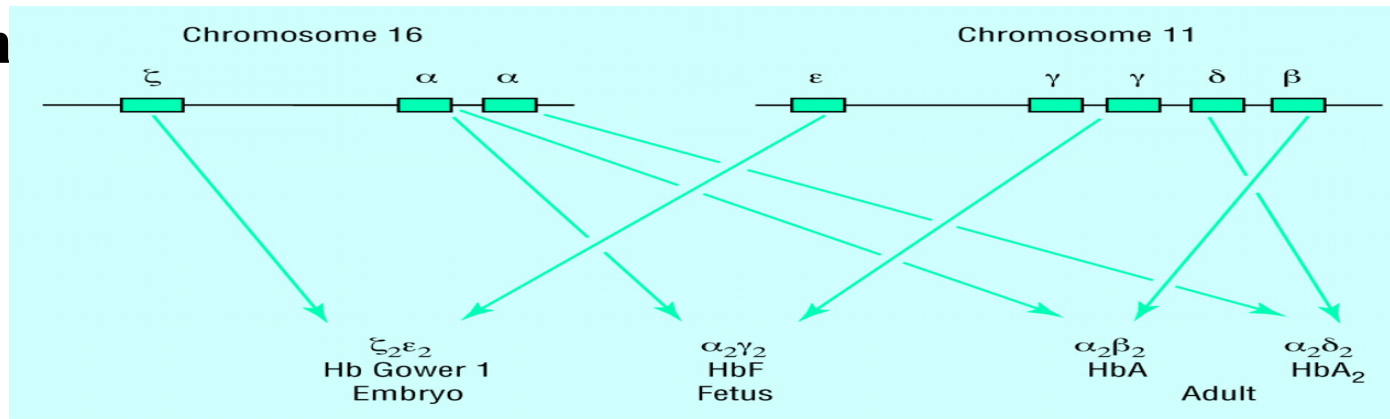
- Chromosomal distribution of the human genes for**
- the α family of globins on chromosome 16,
 - the β family of globin on chromosome 11

Gene structure is shown by black bars (exons) and colored bars (introns).

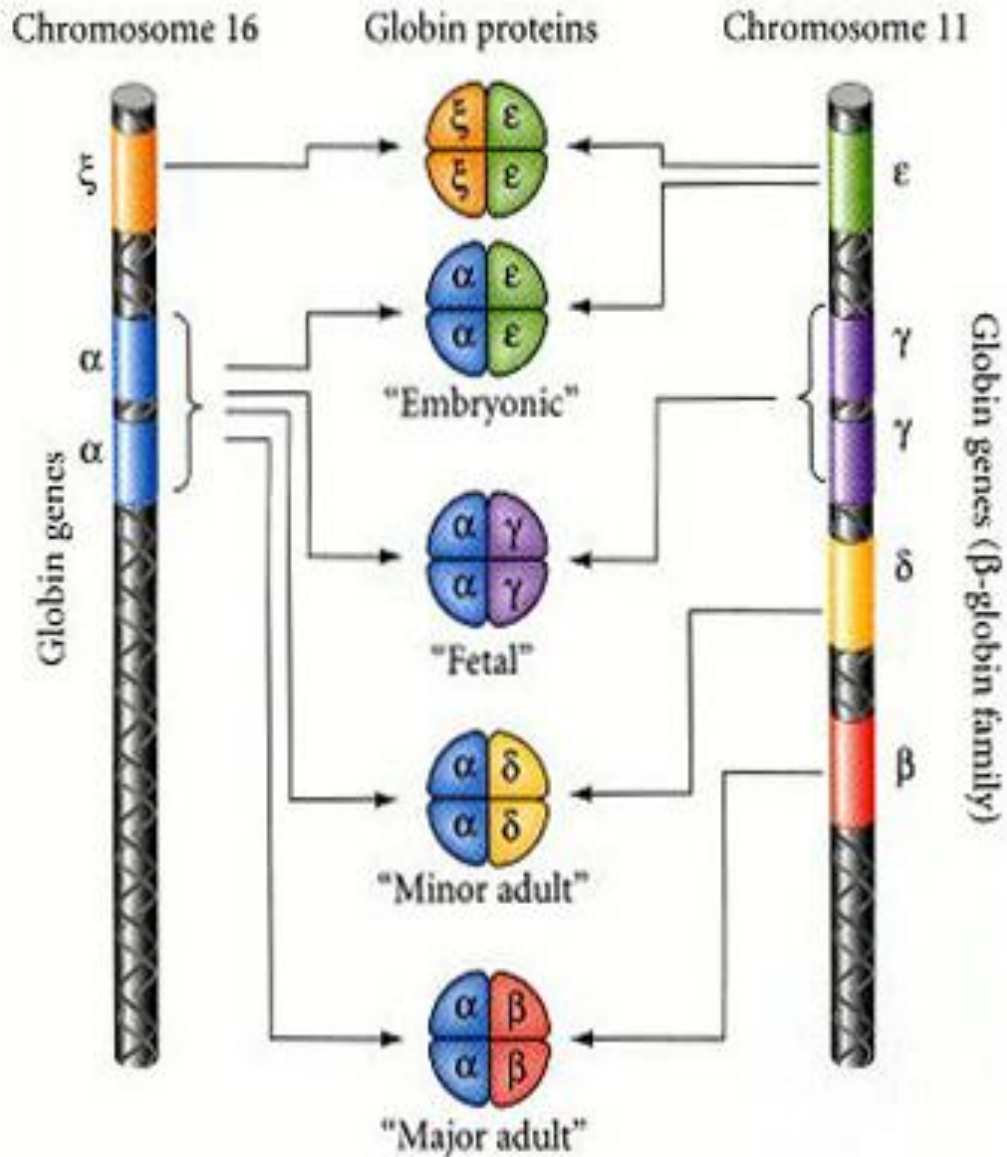


ORGANIZATION OF GLOBIN GENE FAMILY IN HUMAN

- The **b**, **d**, **g**, and **e** chains all belong to an "b-like" group; they have very similar amino acid sequences and are encoded by genes of identical intron-exon structure that are all contained in a 60-kb stretch of DNA on chromosome 11.
- The **a** and **z** chains belong to an "a-like" group and are encoded by genes contained in a 40-kb region on chromosome 16. Two slightly different forms of the **a** chain are encoded by neighboring genes with identical intron



GLOBIN GENES



The various forms of Hb molecules and the genes from which they are coded

α -Chain

- Maps to chromosome 16

- Polypeptide length of 141 AA

β -Chain

- Maps to chromosome 11

- Polypeptide length of 146 AA

IV. NORMAL HEMOGLOBIN

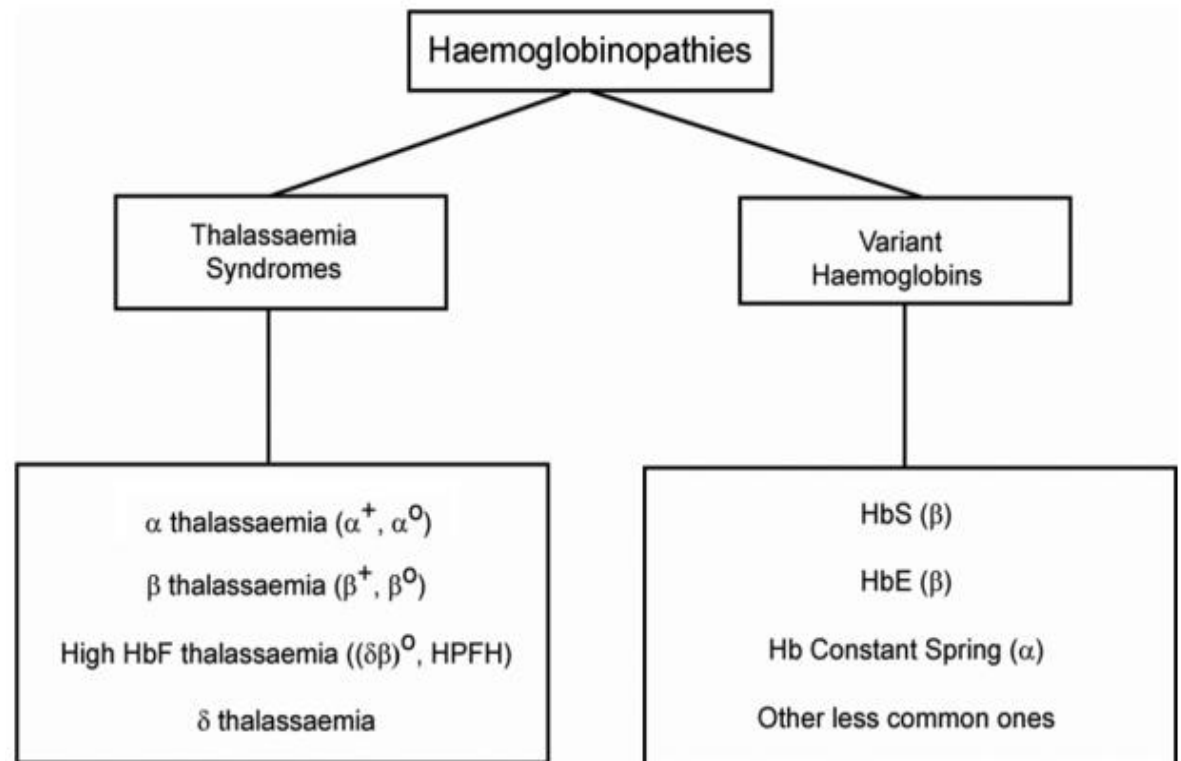
DEVELOPMENTS OF HEMOGLOBIN

- In the first 8 weeks of embryonic life the predominant forms of Hb are: - **Hb Gower 1 ($\zeta_2\varepsilon_2$)**.
 - **Hb Gower 2 ($\alpha_2\varepsilon_2$)**.
 - **Hb Portland 1 ($\zeta_2\gamma_2$)**.
- By the 12th week embryonic Hb is replaced by **Hb F ($\alpha_2\gamma_2$)** which represents 70–100% of Hb in fetal life.
- Adult Hb (**Hb A ($\alpha_2\beta_2$)**) detectable from 16/40, replaces Hb F as predominant Hb by 6/12 after birth, up to 30% of Hb in fetal life.
- **HbA2 ($\alpha_2\delta_2$)** is present in utero, but only very minor in normal adults.
- In normal adults 96–98% of Hb is HbA, Hb A2 (2–3%)

ABNORMAL HEMOGLOBIN

Hemoglobinopathies

- Thalassemia genetics
 - Hb synthesis
 - Hb A, A₂, F
 - Hb Constant-Spring
 - Hb Bart's
 - Hb ELP
 - Hb H
 - Hb Lepore
 - Hb E
- Hb S
 - Hb C
 - Hb SC disease
 - HPFH



CLASSIFICATION OF HEMOGLOBINOPATHY

I. Thalassaemia:

- **Primary abnormality is a reduced rate of synthesis of globin chains. Imbalance α - β chain ratio.**
- **Hemoglobin synthesis characterized by the absence or reduced amount of one or more of the globin chains of hb.**
 - α -thalassemia
 - β -thalassemia

II. Abnormal Hb with amino acid changes.

- **There is a structural abnormality in the globin chain.**
- **In sickle cell disease, a missense mutation (glu $\rightarrow$ val at codon 6) replaces a polar by a neutral amino acid on the outer surface of the β -globin molecule.**
- **Other amino acid changes can cause anemia, cyanosis,**

I. THALLASSEMIA

History

- **Thomas Cooley (1925):** described a severe type of anemia in children of Italian origin: → the erythroblastic anemia and developed splenomegaly during the first years of life, usually before age 10 → so named Cooley's anemia.
- *Riethe and Wintrobe in Europe (1925): a Italian children with unexplained **mild hypochromic and microcytic anemia**.*
- **In 1936:** all disorders designated diversely as von Jaksch's anemia, splenic anemia, Cooley's anemia, erythroblastosis, and Mediterranean anemia, were in fact a single entity, mostly seen in patients who came from the Mediterranean area, the disease name is 'thalassemia' derived from the Greek word, meaning 'the sea'. It was also recognized that Cooley severe anemia was the **homozygous form** of the **mild anemia** described by Riethe and Wintrobe. The severe form then was labeled as **thalassemia major** and the mild form as **thalassemia**

Thalassemias are a heterogenous group of genetic disorders.

- **Individuals with homozygous forms are severely affected and **die early** in childhood without treatment**

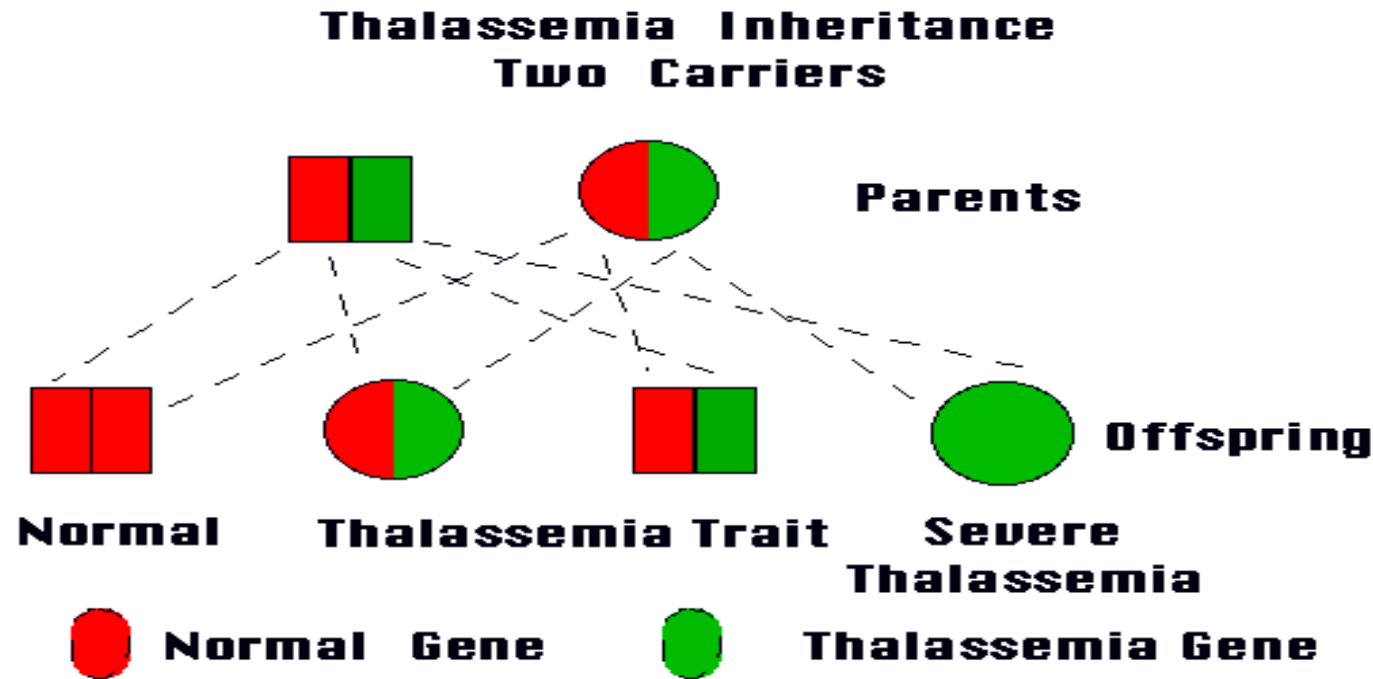
- *Heterozygous individuals exhibit varying levels of severity*

The disorders are due to mutations that decrease the rate of synthesis of one of the 2 globin chains (α or β).

The genetic defect may be the result of:

- **A mutation in the noncoding **introns** of the gene resulting in inefficient RNA splicing to produce mRNA, and therefore, **decreased mRNA production****
 - **The **partial** or total deletion of a globin gene**
 - **A mutation in the **promoter** leading to decreased expression**
 - **A mutation at the **termination** site leading to production**
- 

THALASSEMIA INHERITANCE



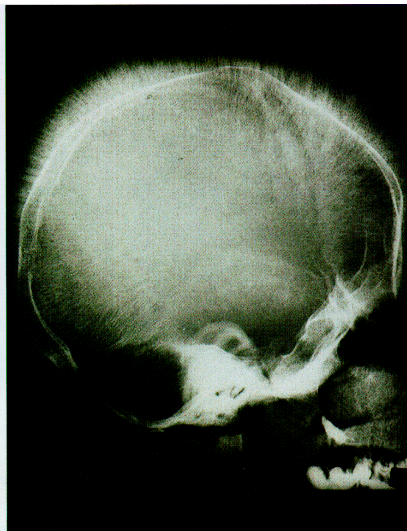
Named after globin chain that is abnormally synthesized

- Reduced or absent α -globin chain : α -thalassemia
- Reduced or absent β -globin chain : β -thalassemia
- Reduced or absent γ -globin chain : γ -thalassemia
- Reduced or absent δ -globin chain : δ -thalassemia
- Reduced or absent ϵ -globin chains:
gdb-thalassemia

Hb A	$\alpha_2\beta_2$	96%
Hb A ₂	$\alpha_2\delta_2$	3%
Hb F	$\alpha_2\gamma_2$	1%

Beta Thalassemia	Delta-Beta Thalassemia	Alpha Thalassemia
$\alpha_2\beta_2\downarrow$	$\alpha_2\beta_2\downarrow$	$\downarrow\alpha_2\beta_2$
$\alpha_2\delta_2$	$\alpha_2\delta_2\downarrow$	$\downarrow\alpha_2\delta_2$
$\alpha_2\gamma_2$	$\alpha_2\gamma_2$	$\downarrow\alpha_2\gamma_2$

THALASSEMIA



**Dark skin
due to iron
overload**



1. ALPHA-THALASSEMIA

- In α -thalassemia is deficient synthesis of α -chains.
- The resulting excess of β -chains bind oxygen poorly, leading to a low concentration of oxygen in tissues (hypoxemia).
- Deletions of HbA1 and/or HbA2 tend to underlie most cases of α -thalassemia. The severity of symptoms depends on how many of these genes are lost.
- Reduced copy numbers of α -globin genes produce successively more severe effects.

Clasification of α -thalassemia

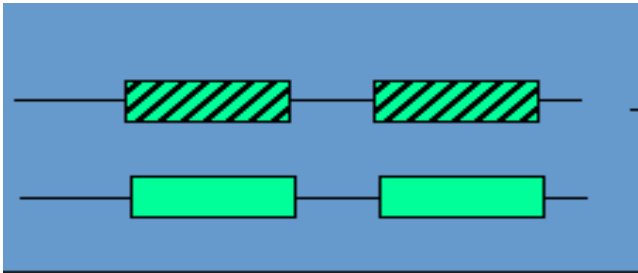
- normal : $\alpha\alpha/\alpha\alpha$
- silent carrier : $-\alpha/\alpha\alpha$
- minor /trait : $-\alpha/-\alpha$ or $--/\alpha\alpha$ → mild microcytic anemia
- Hb H disease : $--/-\alpha$ → severe hemolytic



TYPES OF ALPHA-THALASSEMIA

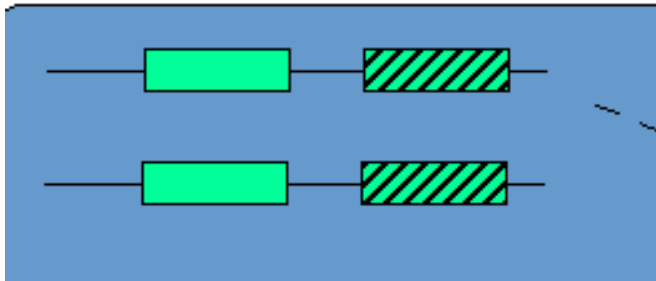
COMPOUND HETEROZYGOTES

α -thalassemia-1 or α^0 -thalassemia ($--$)



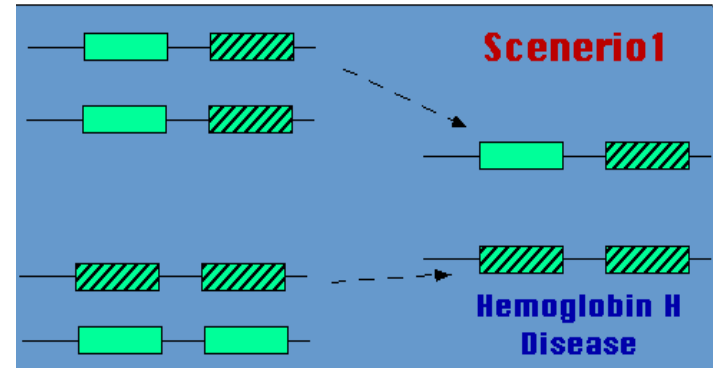
$--/\alpha\alpha$

α -thalassemia-2 or α^+ -thalassemia ($-\alpha$)

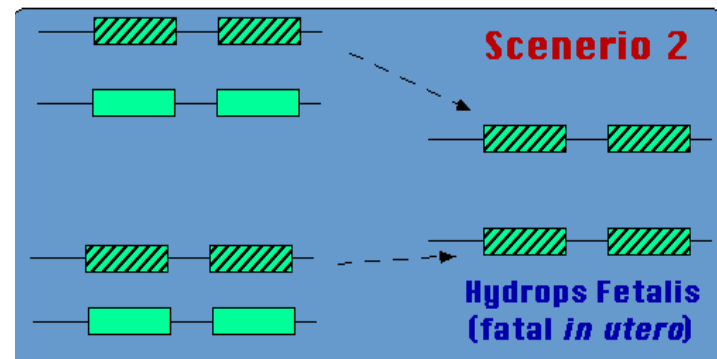


$-\alpha/-\alpha$

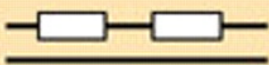
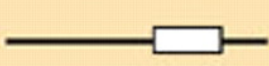
Hb H disease ($--/-\alpha$)



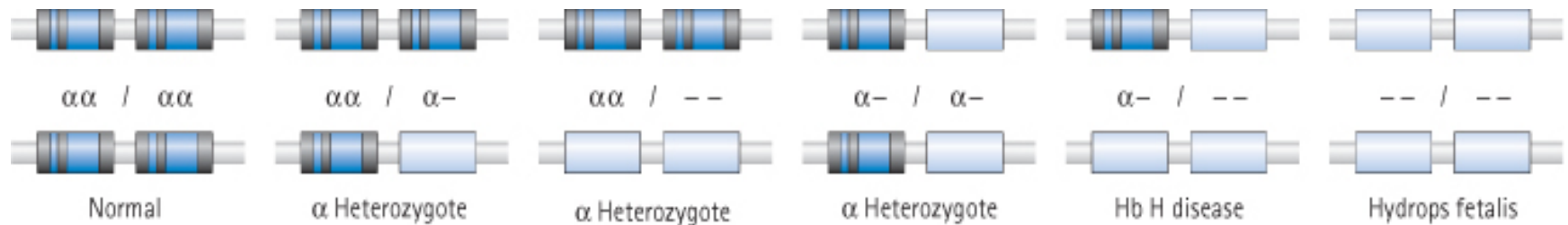
Hb Bart's hydrops fetalis syndrome ($--/--$)



ALPHA-THALASSEMIA SYNDROMES

α -Thalassemia Syndromes		
α Gene Map	α Genotype	α Clinical Syndrome
	Normal	Normal
	Heterozygous α - Thal - 2 (also called α^+)	Silent Carrier of α Thalassemia
	Heterozygous α - Thal - 1 (also called α^0)	α - Thalassemia Trait
	Homozygous α - Thal - 2 (also called homozygous α^+)	α - Thalassemia Trait
	Compound Heterozygous α - Thal - 1 & 2 (also called α^+/α^0)	Hb - H Disease
	Homozygous α - Thal - 1	Hydrops Fetalis

VARIOUS FORMS OF ALPHA-THALASSEMIA

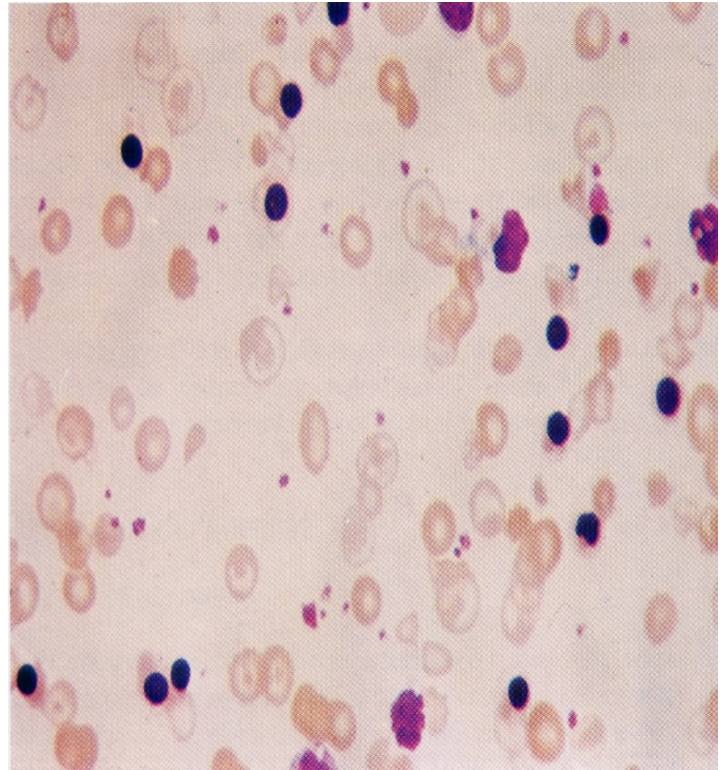
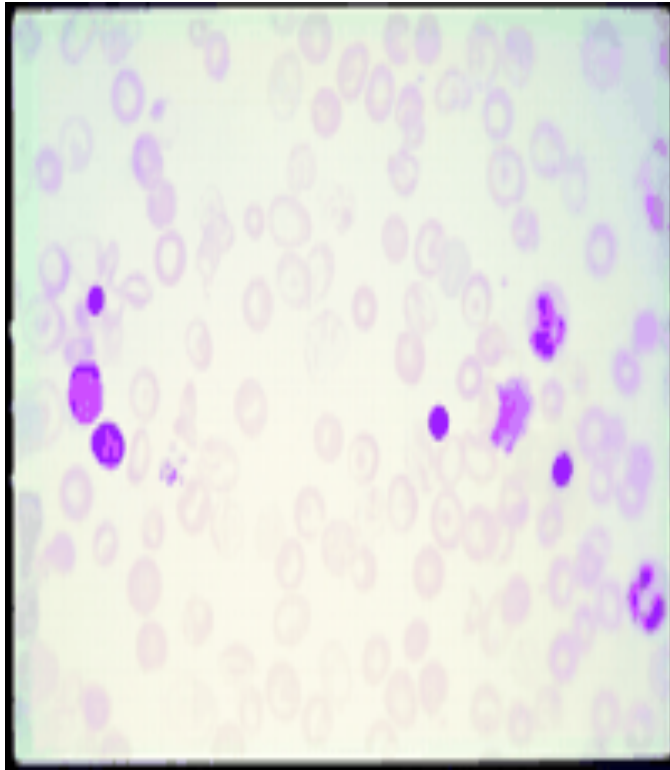


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Thalassemic face



THALASSEMIA MAJOR



Alpha Thalassaemia Major - target cells, microcytosis, hypochromia, NRBCs, poikilocytes, polychromasia, basophilic stippling, tear drops.



2. BETA-THALASSEMIA

- **B Thal** are the most important types of thalassemias because they are so common and usually produce severe anemia in their homozygous and compound heterozygous states (compound = when combined with other hemoglobinopathies or thalassemias)
 - **B Thal** are autosomal inherited disorders of β globin synthesis. In most, globin structure is normal but the rate of production is reduced because of decrease in transcription of DNA, abnormal processing of pre-mRNA, or decreased translation of mRNA leading to decreased Hb-A production (A=Adult).
- 

Beta-thalassemia

The disease manifests itself when the **switch from γ to β chain synthesis** occurs several months after birth

There may be a **compensatory increase in γ and δ chain synthesis** resulting in increased levels of hb F and A2.

The genetic background of β thal is heterogenous and may be roughly divided into two types:

β^0 in which there is **complete absence** of β chain production. This is common in the Mediterranean.

β^+ in which there is a **partial block** in β chain synthesis.

At least three different mutant genes are involved:

β^{+1} – 10% of normal β chain synthesis occurs

β^{+2} – 50% of normal β chain synthesis occurs

β^{+3} - > 50% of normal β chain synthesis occurs



In β^+ and β^{++} thal, the mutated gene encodes for a small amounts of normal β mRNA. The quantity of β globin chain, which are made, varies largely from one molecular mutation/ defect to another.

β thalassemia may also be found in combination with any of the hemoglobinopathies (S, C, or E) leading to a mild to severe anemia depending upon the particular combination.

Hereditary persistence of hb F: are a group of heterogenous disorders with the absence of δ and β globin chains synthesis
→ compensated for by an increase in γ chain synthesis
leading to 100% hgb F.

→ increased affinity for O₂, this results in polycythemia.



CLASIFICACION OF BETA-THALASSEMIA

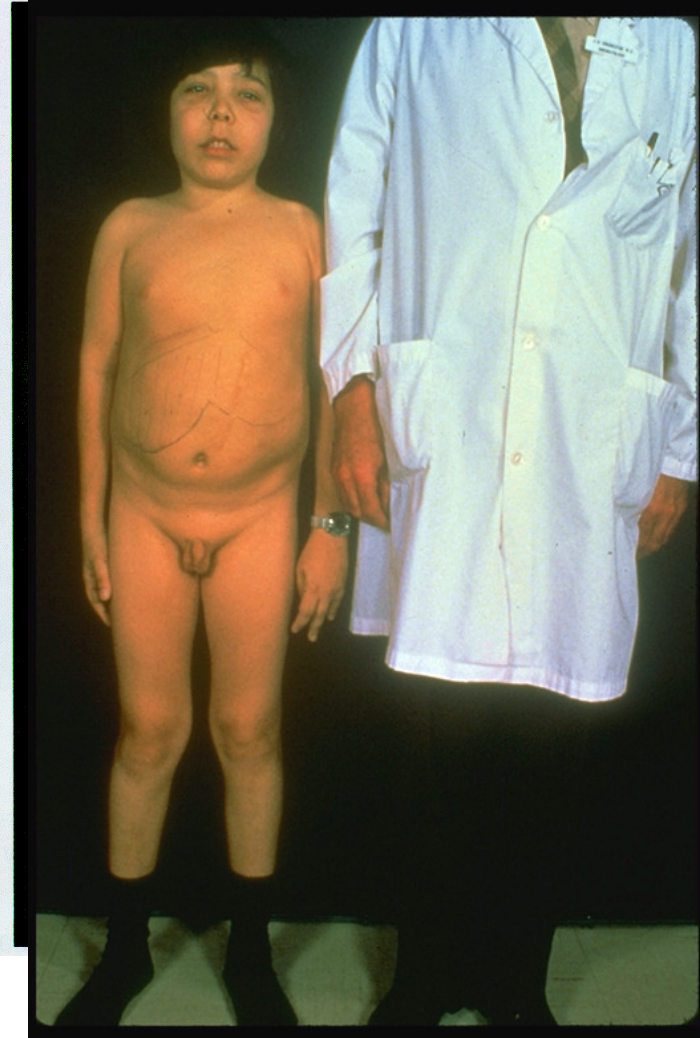
- β thal major is homozygosity or compound heterozygosity resulting in severe phenotype.
- β thal minor or trait is heterozygosity with asymptomatic phenotype.
- β thal intermedia is an intermediate phenotype produced by a variety of genotypes.
- β^0 syndromes are characterized by the affected gene producing no beta chain.
- β^+ syndromes are characterized by the abnormal gene producing beta chains at a reduced rate.
- Usually due to point mutations.



SUMMARY OF GENETIC DEFECT BETA-THALLASSEMIA

- β^+ : reduced beta-globin chain synthesis
 - β^0 : no beta-globin chain synthesis
 - > 150 point mutations and several deletional mutations in beta-globin chain gene all affecting the **expression of the beta-globin chain gene resulting in defects in activation, initiation, transcription, processing, splicing, cleavage, translation, and/or termination**
 - genetic defect:
abnormal or no synthesis of the beta-globin chain → bone marrow fails to produce adequate erythrocytes and increased hemolysis of circulating erythrocytes → anemia → medullary hematopoiesis and extramedullary hematopoiesis (hepatosplenomegaly, lymphadenopathy)
- 

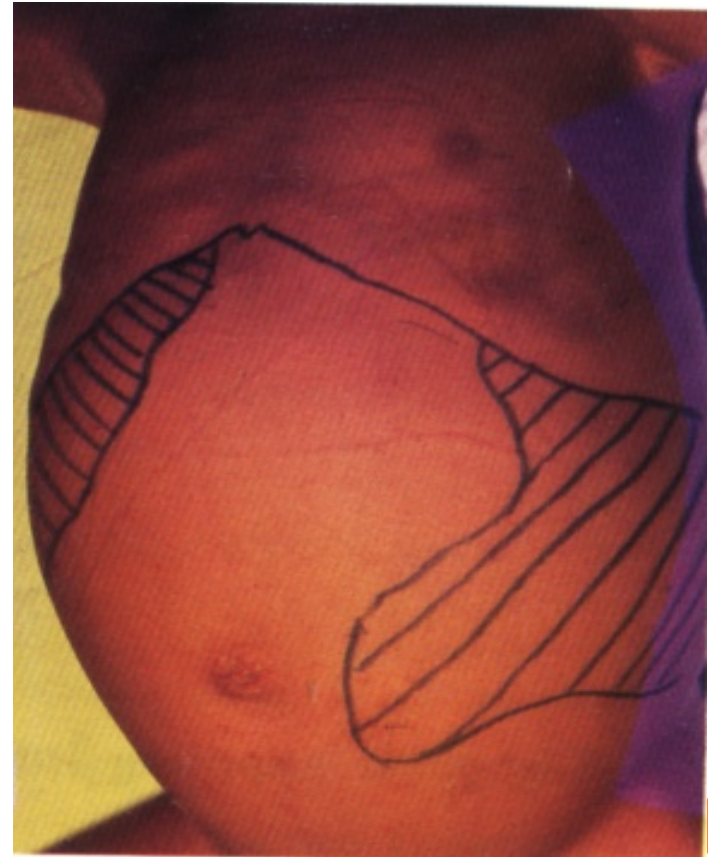
BETA-THALASSEMIA MAJOR



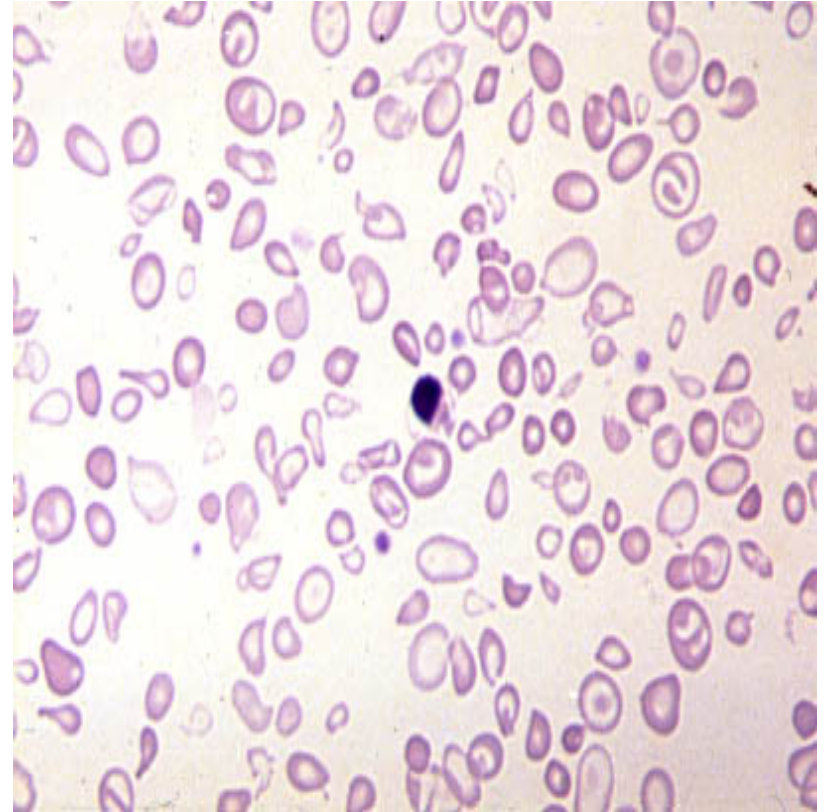
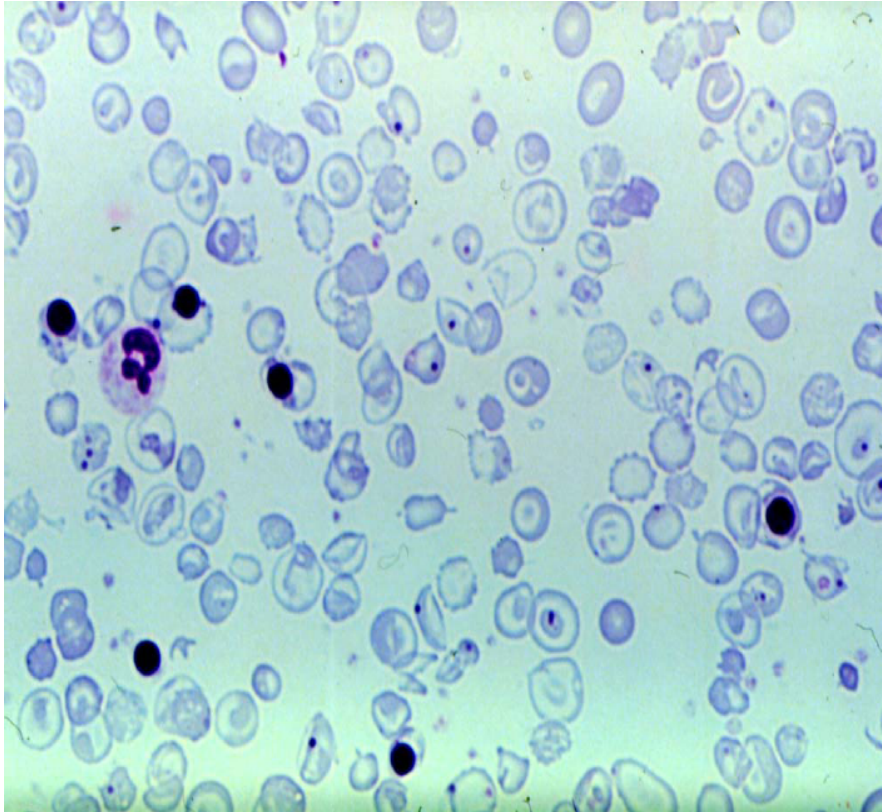
BETA-THALASSEMIA MAJOR



Hepatosplenomegaly



Beta-Thalassemia Major



Anisopoikilocytosis, anisocytosis, microcytosis, hypochromia, fragments, tear drops



Beta-Thalassemia Minor (Trait)

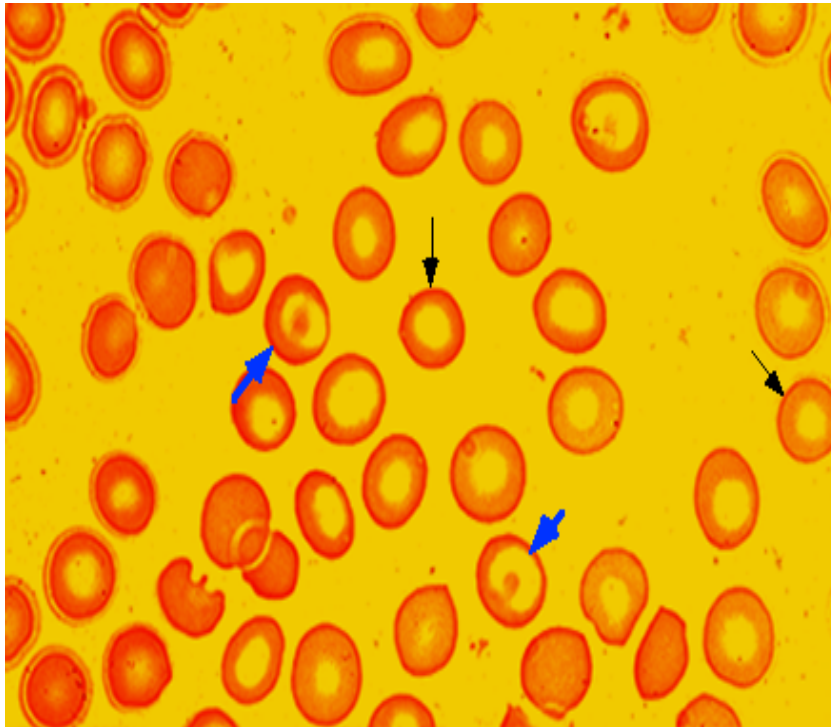
- Caused by heterozygous (from one parent) mutations that affect β globin synthesis.
- β Chains production and thus Hb-A production is more reduced than the silent carrier Hb-A.
- Usually presents as mild, asymptomatic hemolytic anemia → anemia hypochromic microcytic
- Have one normal β gene and one mutated β gene.
- Hb level in 10-13 g/dL range with normal or slightly elevated RBC count (RCC).
- Have high HbA2 levels (3.6-8.0%) and normal to slightly

elevated HbF levels: - 2-6% HbF (N = <1% after age 1 year)

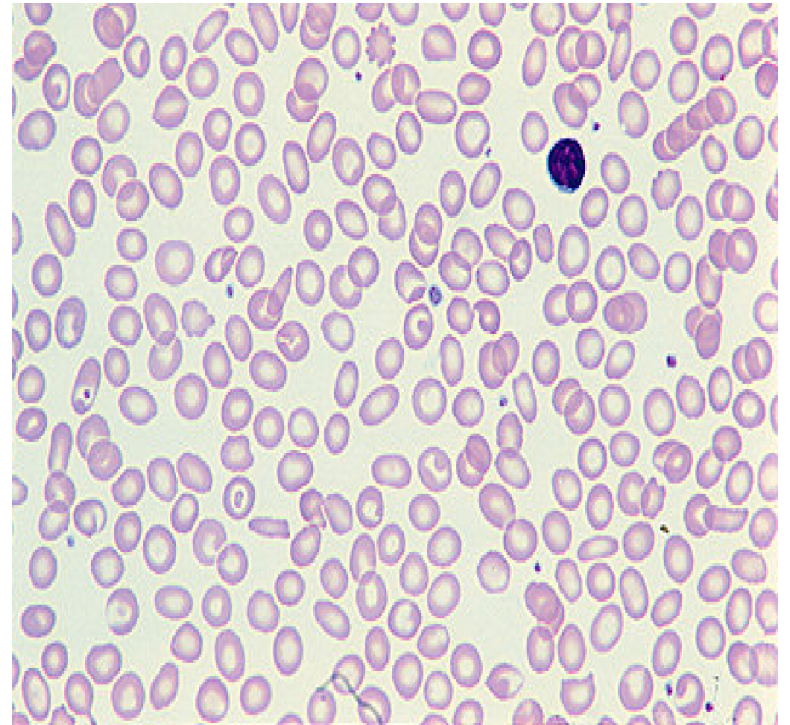
- 6-8% HbA2 (N = 2.2-3.6%)

- 87-95% HbA (N=95-100%)

BETA-THALASSEMIA MINOR



Beta thalassemia trait Peripheral smear from a patient with beta thalassemia trait. The field shows numerous hypochromic and microcytic red cells (thin arrows), some of which are also target cells (blue arrows). Courtesy of Stanley Schrier, MD



B-THAL → THE CLINICAL EXPRESSION

The clinical expression of the different gene combinations (1 from mom and 1 from dad) are as follows:

β^0/β^0 , β^{+1}/β^{+1} , or $\beta^0/\beta^{+1,+2}$, or $+3$ = thalassemia major
→ decreased total RBC: **hypochromic, microcytic** anemia.

α chains may also precipitate leading to pitting in the spleen and **decreased RBC survival** via a chronic hemolytic process.

β^{+2} , or 3 homozygous = thalassemia intermedia

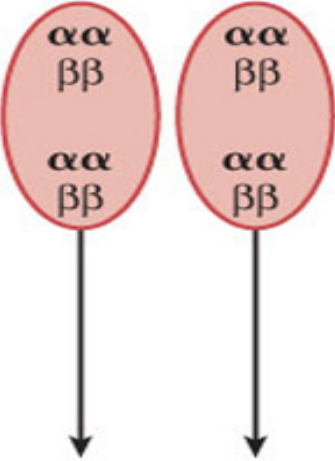
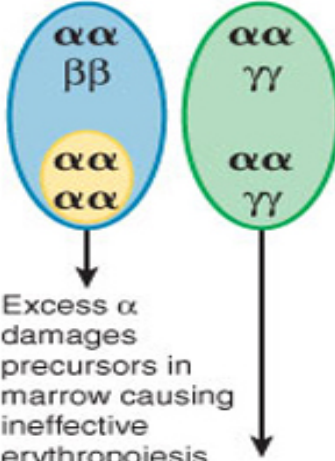
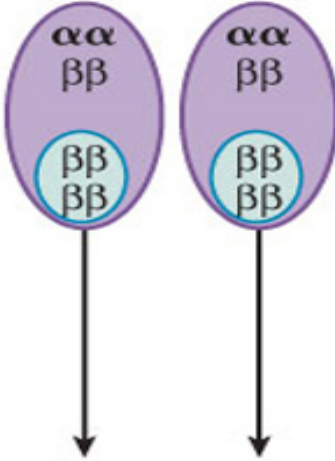
Heterozygosity of β^0 , or β^{+} = thalassemia minor

→ mild hypochromic, microcytic anemia

δ/β thalassemia: **both δ and β chains are absent with no or little compensatory increase in γ chain synthesis. This leads to 100% hgb F and mild hypochromic, microcytic anemia**



ALPHA- AND BETA-THALASSEMIAS

	Normal	Severe β thalassemia	Severe α thalassemia (hemoglobin H disease)
Globin production	Balanced α and β production	Decreased β production	Decreased α production
Bone marrow		 Excess α damages precursors in marrow causing ineffective erythropoiesis	
Peripheral blood	Normal hemoglobin composition (97% hemoglobin A)	Selective survival of cells producing hemoglobin F	Hemoglobin H detectable in blood. Hemoglobin H precipitates and damages RBCs, causing hemolysis

COMPARISON OF ALPHA-THALASSEMIAS

Phenotype	Hb A	Hb Barts	Hb H
Normal	97-98%	0	0
Silent Carrier	96-98%	0-2% (At birth)	0
α Thalassemia Trait	85-95%	2-8% (At birth)	<2%
Hb H Disease	Dec	<10% (At birth)	5-40%
Hydrops Fetalis	0	70-80% (with 20% Hb Portland)	0-20%

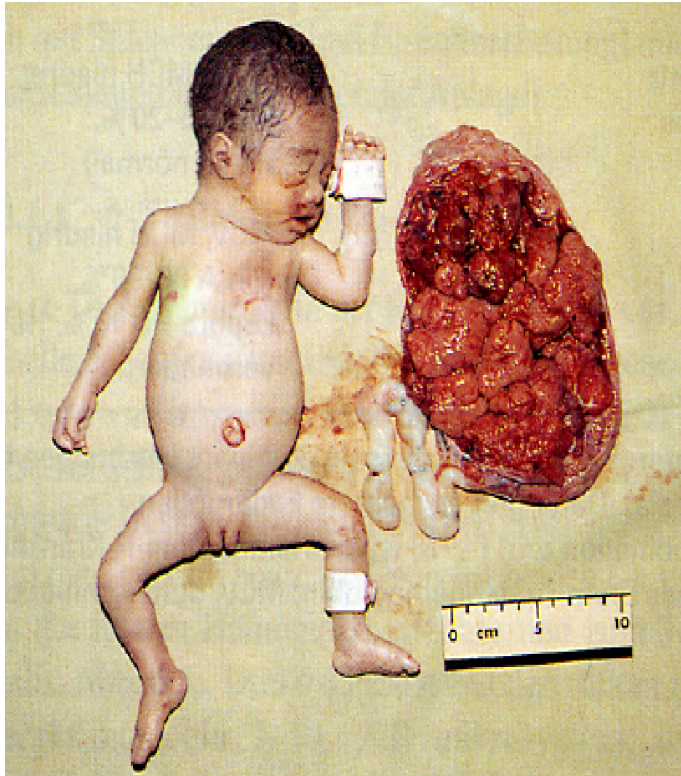


Bart's Hydrops Fetalis Syndrome

- Most severe form. Incompatible with life. Have no functioning α chain genes (- -/- -).
- Baby born with hydrops fetalis, severe anemia, will see hepatosplenomegaly and cardiomegaly.
- Predominant Hb is Hb Bart, along with Hb Portland and traces of HbH.
- Hb Bart's has high oxygen affinity so cannot carry oxygen to tissues.
- Fetus dies in utero or shortly after birth. At birth, you will see severe hypochromic, microcytic anemia.



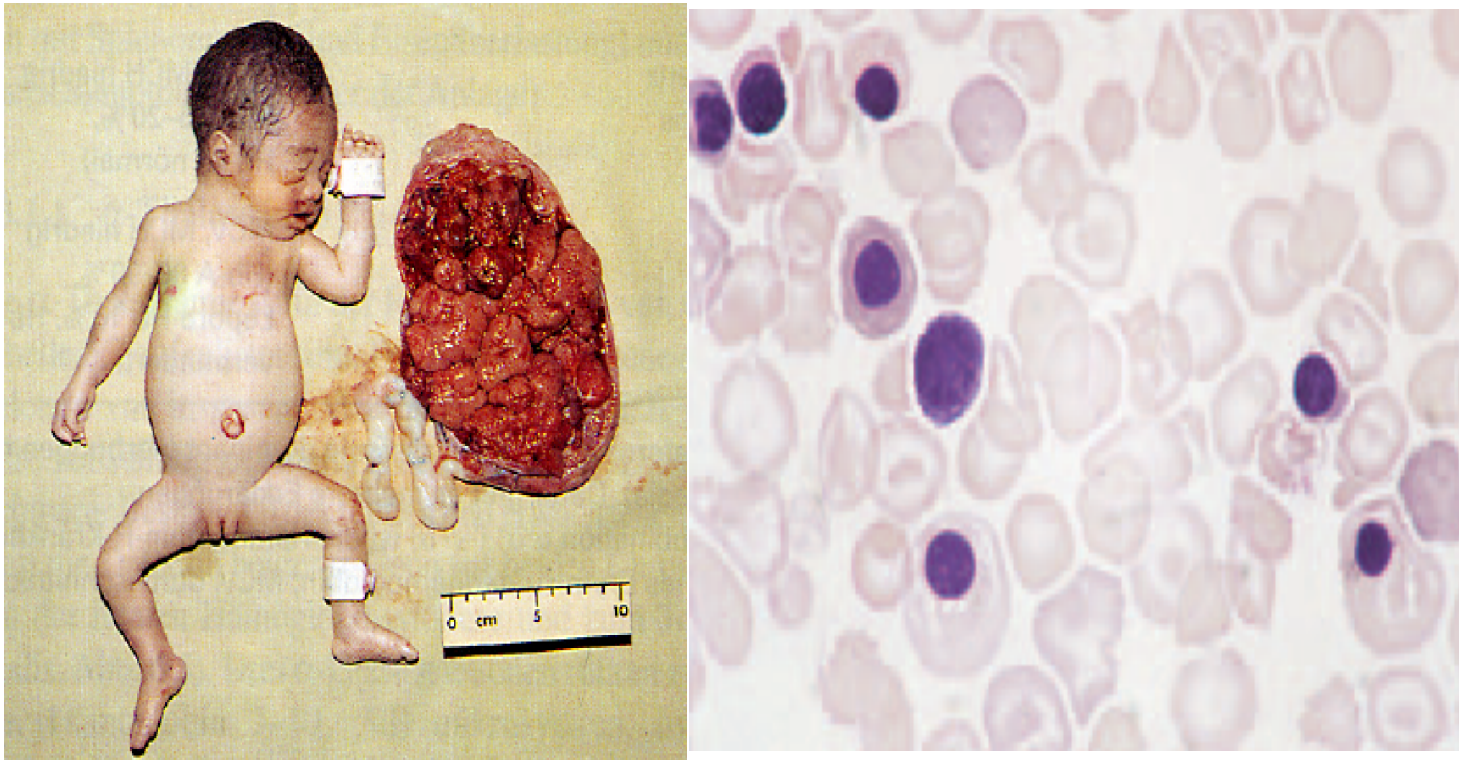
Hydrops Fetalis



Neonatus with hemoglobin Bart's hydrops fetalis
→ anisocytosis, poikilocytosis and erythrocyte with
some nucleus



Hydrops Fetalis



The blood film of neonate with hemoglobin Bart's hydrops fetalis showing anisocytosis, poikilocytosis and numerous nucleated red blood cells (NRBC).



HEMOGLOBIN H DISEASE

- ❑ Second most severe form alpha thalassemia.
- ❑ Usually caused by presence of only one intact α gene producing alpha chains ($--/-\alpha$).
- ❑ Results in accumulation of excess unpaired β or δ chains

Born with 10-40% Bart's hemoglobin (γ_4).

Gradually replaced with Hemoglobin H (β_4).

In adult, have about 5-40% HbH.

γ_4

β_4

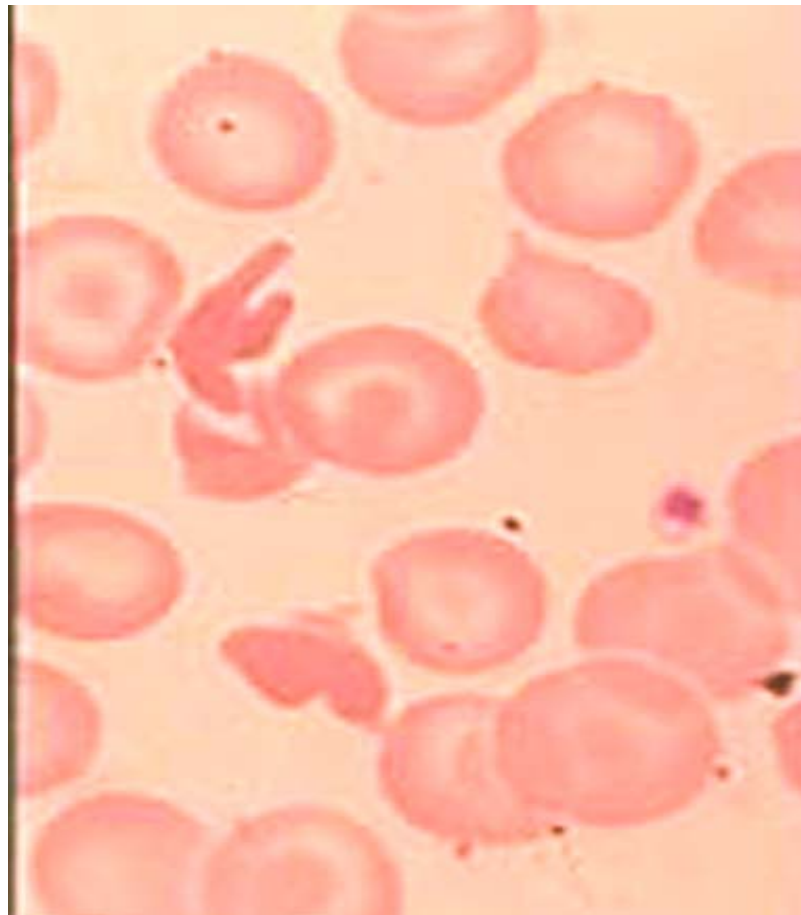
- ❑ **Normal life; however, infections, pregnancy, exposure to oxidative drugs may trigger hemolytic crisis.**
- ❑ **RBCs are microcytic, hypochromic with marked poikilocytosis**
- ❑ **Hb 7-10 g/dl**



Hb C disease



Hb SC disease



ALPHA-BETA-THALASSEMIA

In some cases:

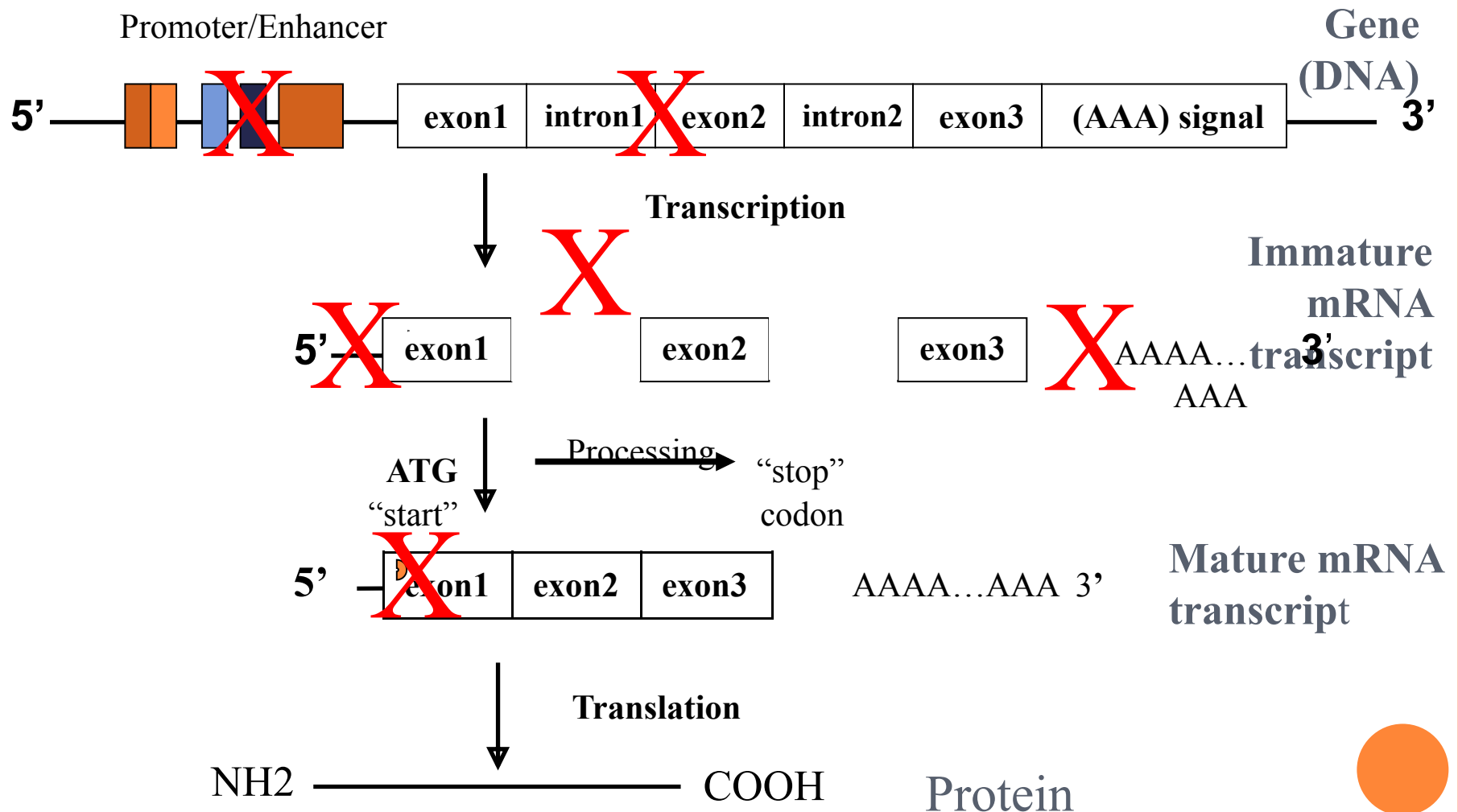
- **They result from deletions and fusion of the δ and β globin genes.**
 1. $\delta\beta^0$ thalassemia: caused by complete deletion of $\delta\beta$ genes. Homozygotes have 100% HbF.
With moderate anemia 8-10 g/dl.
With microcytic and hypochromic.
 2. $\delta\beta^+$ thalassemia: or called Hb Lepore thalassemia.
Homozygotes have only HbF and Hb Lepore.
Heterozygotes have 15-25% Hb-F.
- **In $\delta\beta$ thalassemia not all cells are F cells.
So the distribution of HbF is heterocellular.
This means not all cells are F cells.**



Table 2. Types of haemoglobins and globin chains present in normal adult blood and in haemoglobinopathies

Haemoglobin (Hb)	Globin chains	Percentage found in normal adult blood	Clinical state
HbA	$\alpha^2\beta^2$	~97%	Normal
HbA ₂	$\alpha^2\Delta^2$	2–3%	Normal
HbF	$\alpha^2\gamma^2$	<1%	Normal
			
HbH	β^4	0%	α -thalassaemia
HbBarts	γ^4	0%	α -thalassaemia
	α chain aggregates – insoluble	0%	β -thalassaemia
HbS	$\alpha^2\beta s^2$	0%	Sickle cell disease

Type of mutations that could occur



THERE ARE >400 STRUCTURAL VARIANTS OF NORMAL HB.

THE 4 MOST COMMON STRUCTURAL VARIANTS ARE:

- **Hb S (Sickle cell anemia): mutation in β chain: Glu6Val**
→ in Eastern Afrika
- **Hb C: mutation in β chain: Glu6Lys → in Western Afrika**
- **Hb E: mutation in β chain: Glu26Lys → in Asia**
- **Hb M (Methemoglobin): An oxidizing form of Hb containing ferric iron that is produced by the action oxidizing poisons. Non-functional.**
- **HbS is the first variant to be discovered (1949).**
- **Its main reservoir is Central Africa where the carrier rate approximates 20% → *Heterozygous***
approximately 8% of African-Americans
- **Hb SC disease → Washington monument cells**



Hb Constant Spring (severe): mutation in the alpha globin gene produces an alpha globin elongated by 31 amino acids and 2 normal β chains (excess β chains) → the alpha chain protein are unstable

→ (α thal 1 like phenotype)

Homozygous individuals have mild hypochromic microcytic anemia similar to a mild α thal.

- **HbH** (b4, mild): tetramer composed of 4 beta globin chains. HbH forms in people with 3 gene alpha-thal as well as in people with the combination of 2 gene deletion alpha-thal and Hb Constant Spring.

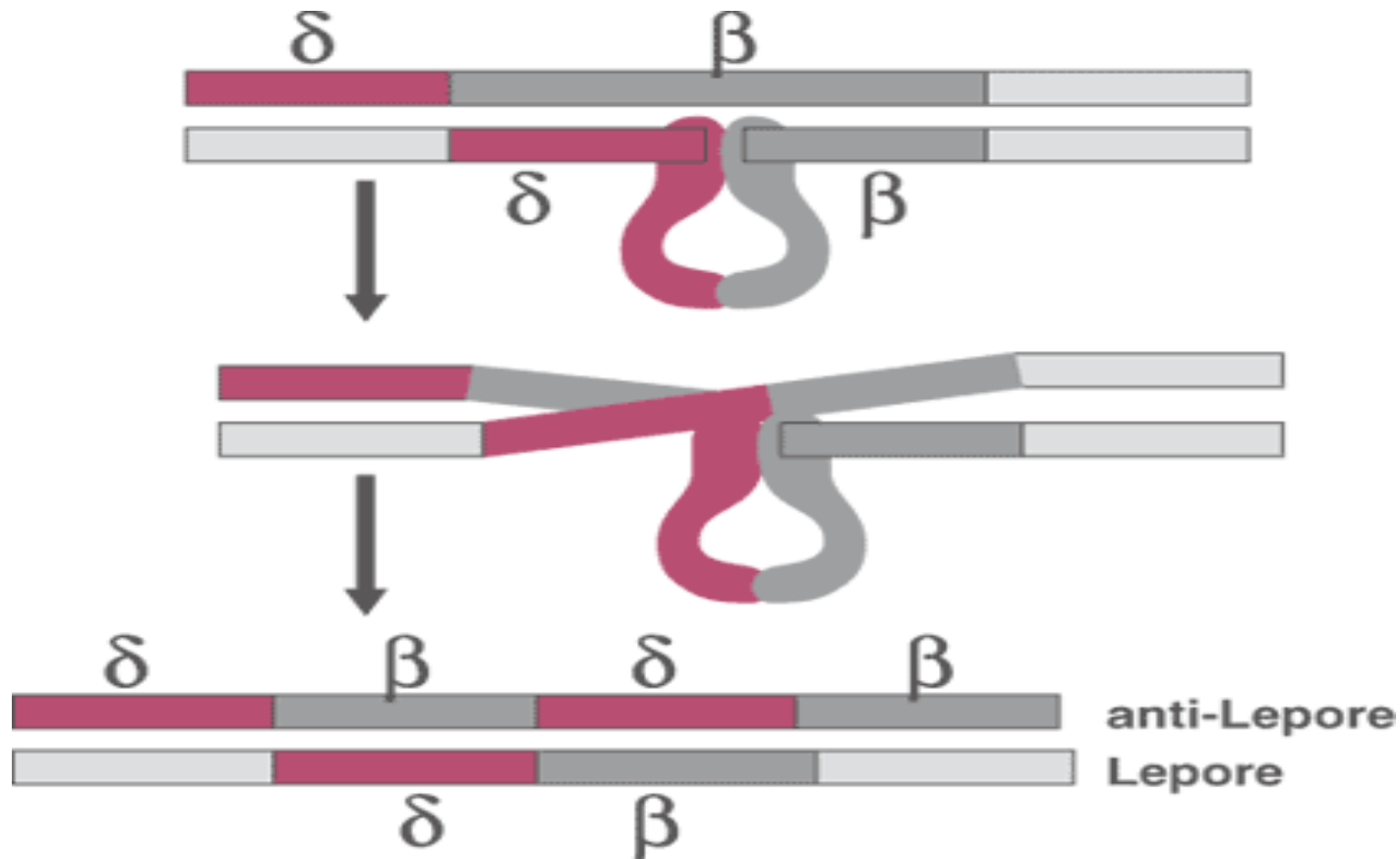
- **Hb Barts** (g4, lethal). With 4 gene deletion alpha-thal no alpha chain is produced. The gamma chains produced during fetal development combine to form gamma chain tetramers.

Individuals with 4 gene deletion and consequent Hb Barts die in utero (hydrops fetalis).



Hemoglobin Lepore: a normal α chain plus a δ - β hybrid (N-terminal δ , and C-terminal β).

There is ineffective synthesis of the hybrid chain leading to α chain excess and the same problems seen in β thalassemia



SOME CLINICALLY SIGNIFICANT VARIANT HEMOGLOBIN

- **HbS (a2bS2, severe).** This the predominant Hb in people with sickle cell disease.
- **HbC (a2bC2, relatively benign).** This results from a mutation in the beta globin gene and is the predominant Hb found in people with hb C disease.
- **HbE (a2bE2, benign).** This variant results from a mutation in the Hb beta chain.
People with HbE disease have a mild hemolytic anemia and mild splenomegaly.



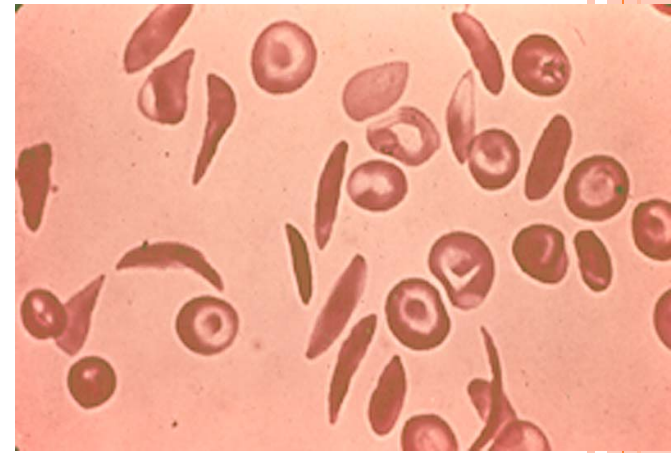
SICKLE CELL ANEMIA (HB S)

The **E6V** (glutamic acid to valine at codon 6) mutation replaces a polar by a neutral amino acid on the outer surface of the b-globin molecule.

Autosomal

The deficiency of oxygen in the blood causes hb S to crystallize, distorting the red blood cells into a sickle shape, making them fragile and easily destroyed → anemia.

Sickled red cells have decreased survival time (→ anemia) and tend to occlude capillaries → the tissue is ischemia and infarction



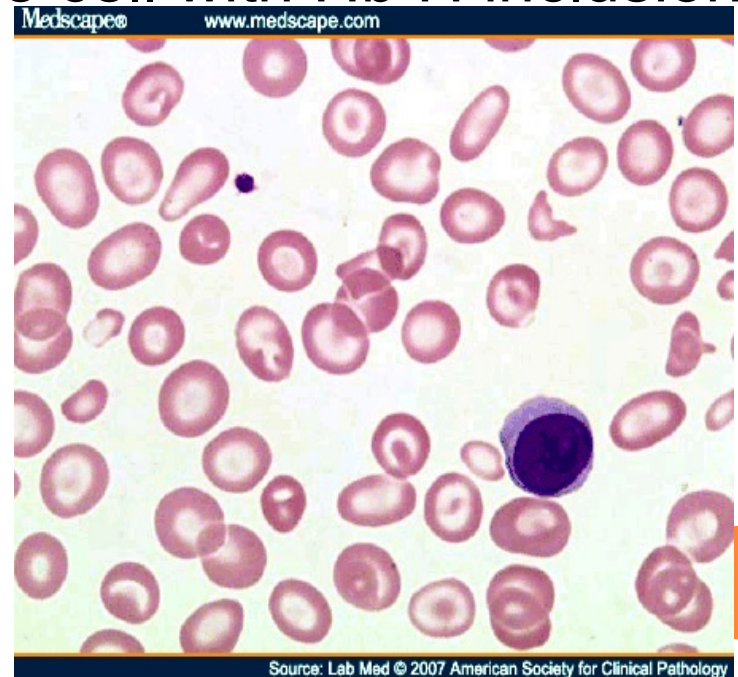
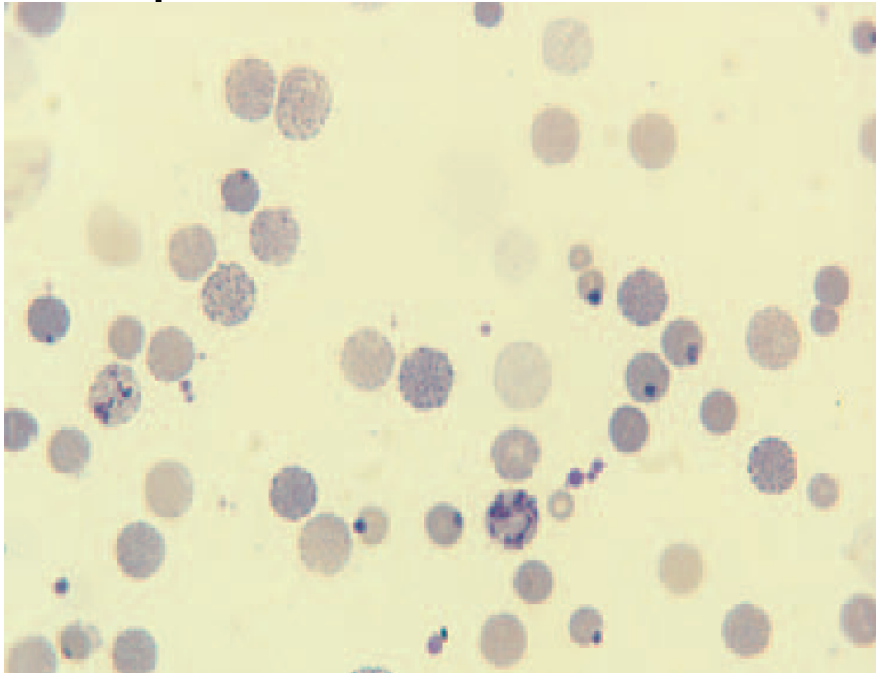
HEMOGLOBIN E

- ✓ Commonly found in Thais
- ✓ Result of point mutation at codon 26 of β -globin gene
- ✓ Glutamic acid→Lysine
- ✓ Phenotype similar to β^+ -thalassemia



HB H INCLUSIONS.

- Hb H is an insoluble tetramer consisting of four beta globin chains, due to a lack of alpha chains in alpha thal major.
- In Hb H disease 30–100% of RBCs contain Hb H inclusions.
- In alpha thal minor there is one cell with Hb H inclusions

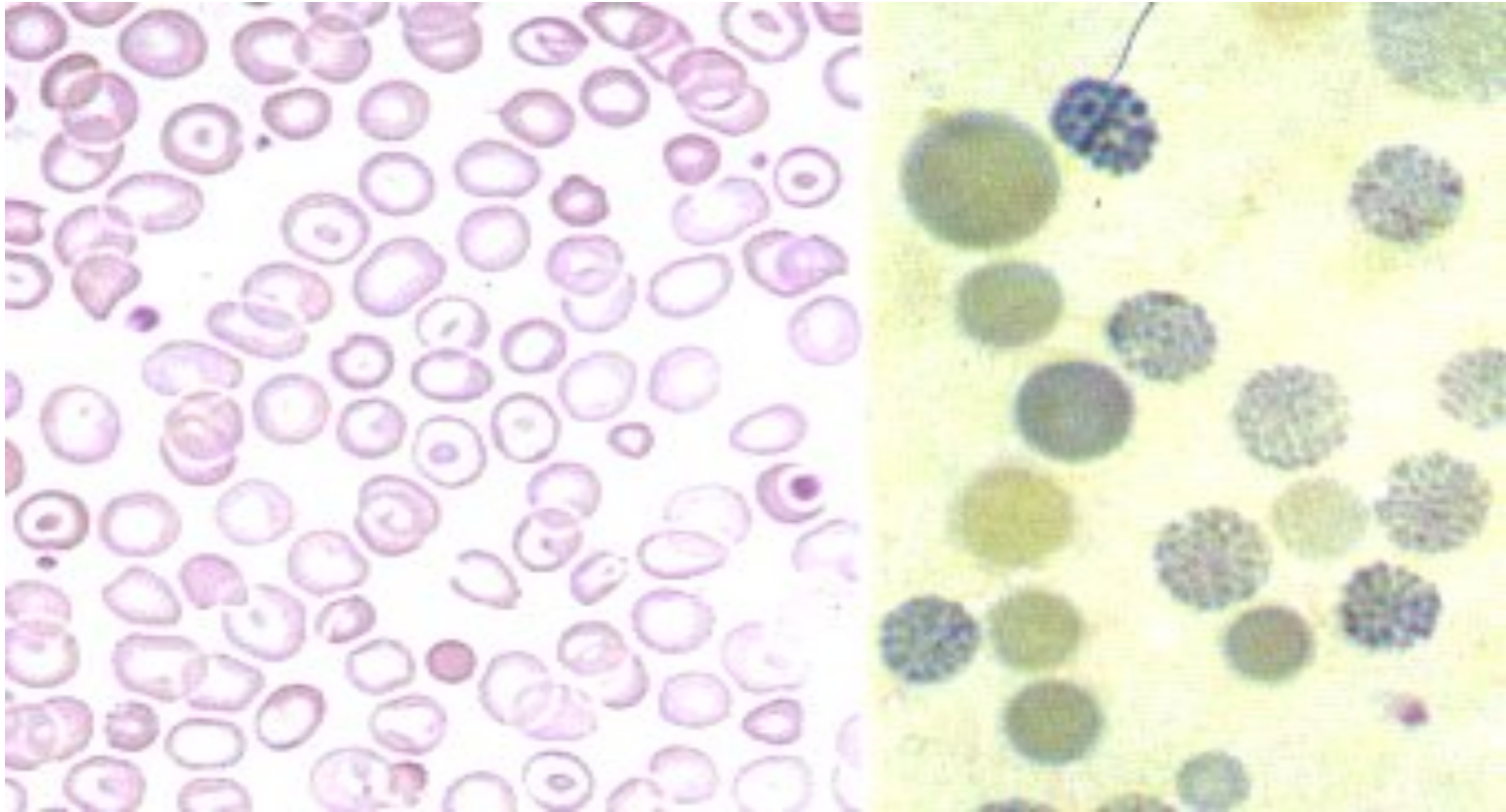


HB CONSTANT SPRING

- Point mutation at termination codon
- UAA → CAA
- 31 amino acids extension
- Total amino acid = 172
- Phenotype similar to α^+ -thalassemia
- Sense mutations involve change from a stop codon to one that codes for an amino acid.
- E.g. Hemoglobin Constant Spring alpha 142 UAA (Stop codon) → CAA (Gln). Translation continues beyond the normal termination until another stop codon (UAA, or UAG, or UGA) is encountered, this causes 31 amino acids elongation. Alpha chain is normally 141 amino acids, but here it is 172 amino acids.



Blood Smear & HbH Preparation



TERIMA KASIH...

MATURNUWUN

